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Wisdom of the Land



Pelvic Fracture For Medical Student

Care in Traumatic Patient (RAID521)

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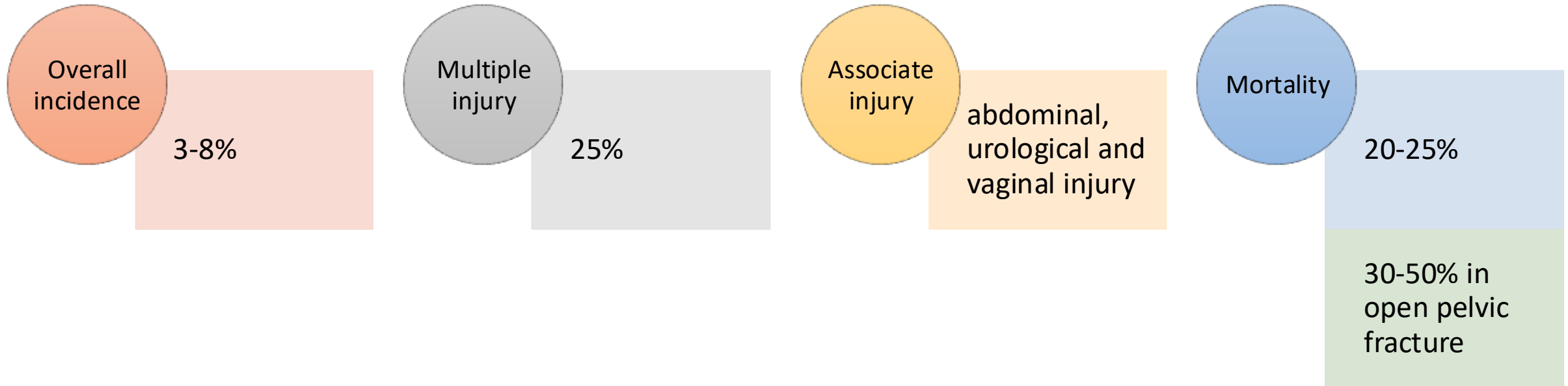
Ramathibodi Medical school

Faculty of medicine Ramathibodi hospital, Mahidol university

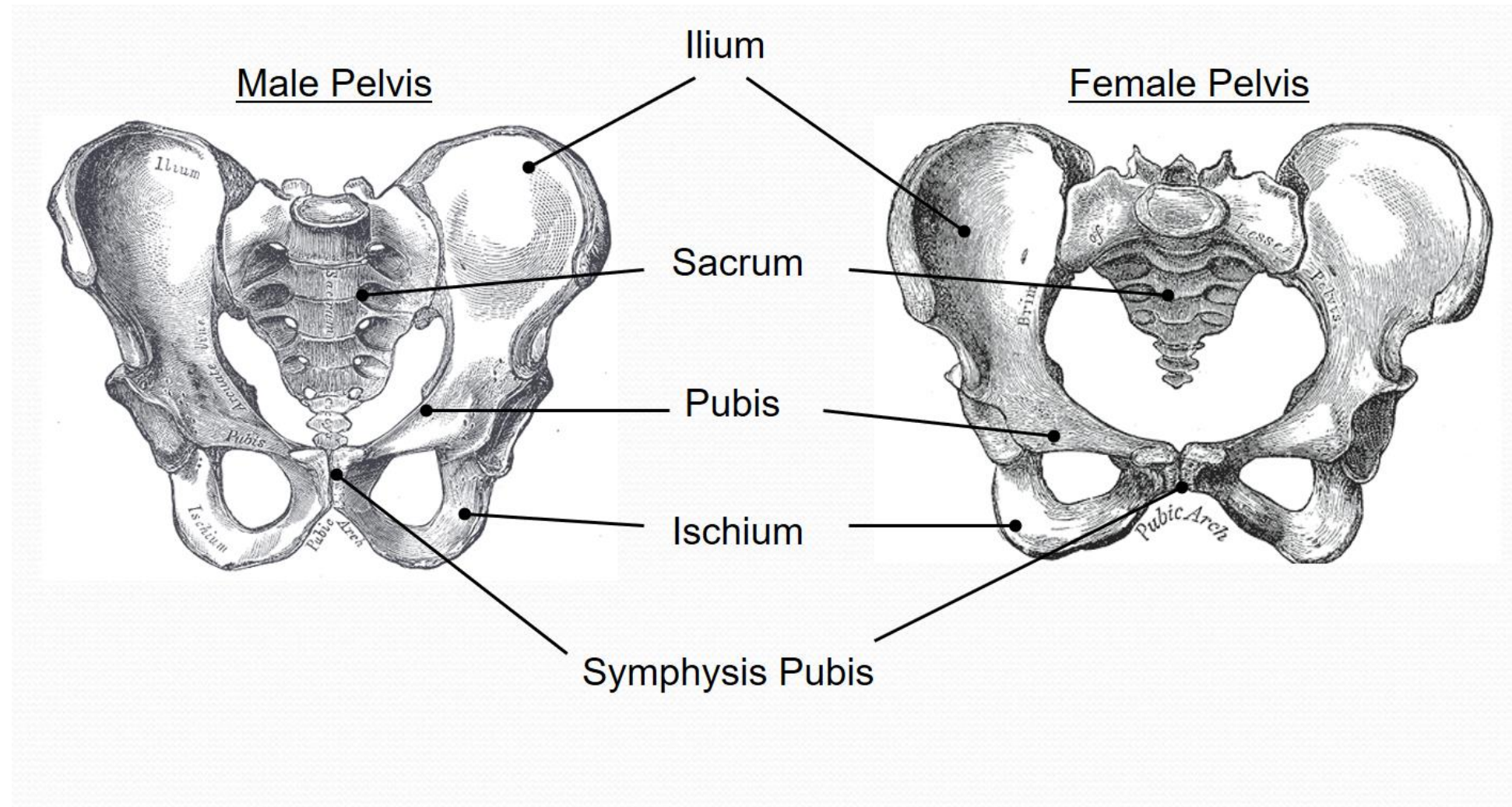
Outline

- Introduction
- Investigation and classification
- Approach to pelvic fracture patient
- Initial management

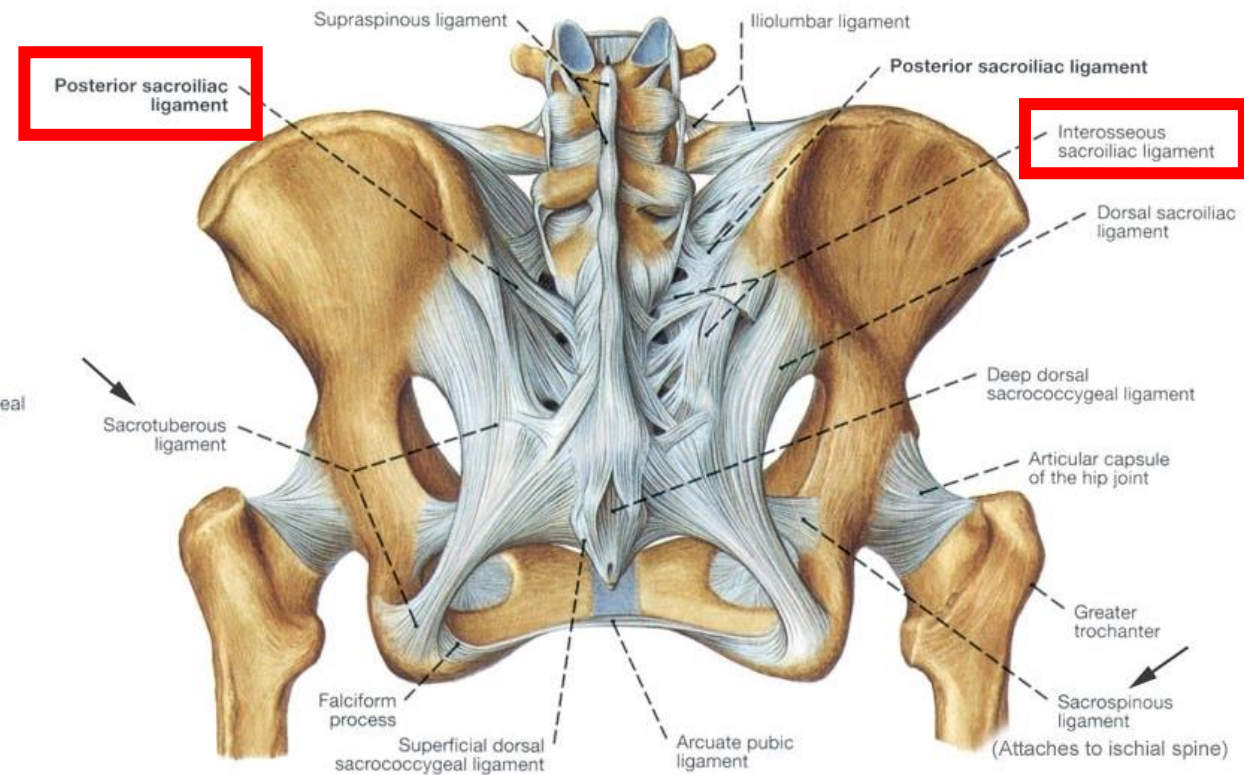
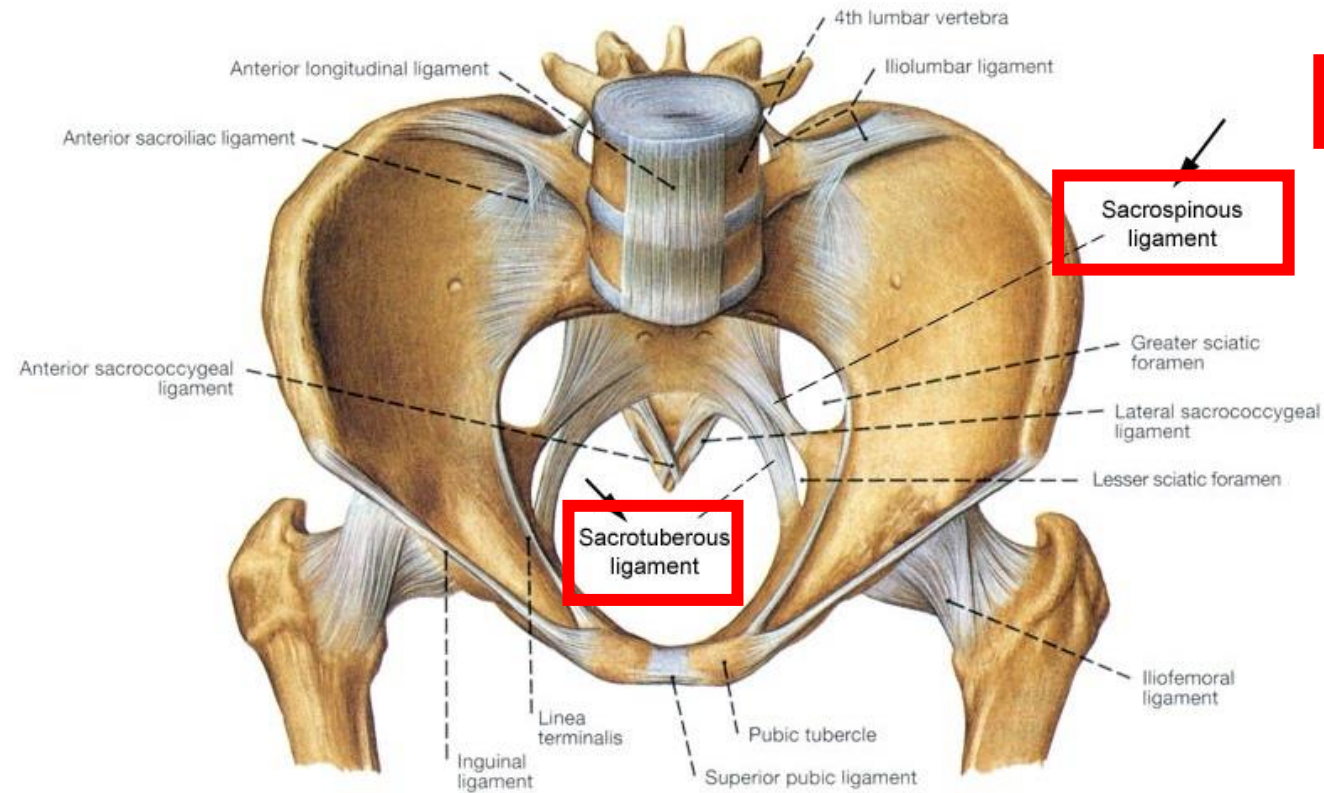
Introduction



Anatomy

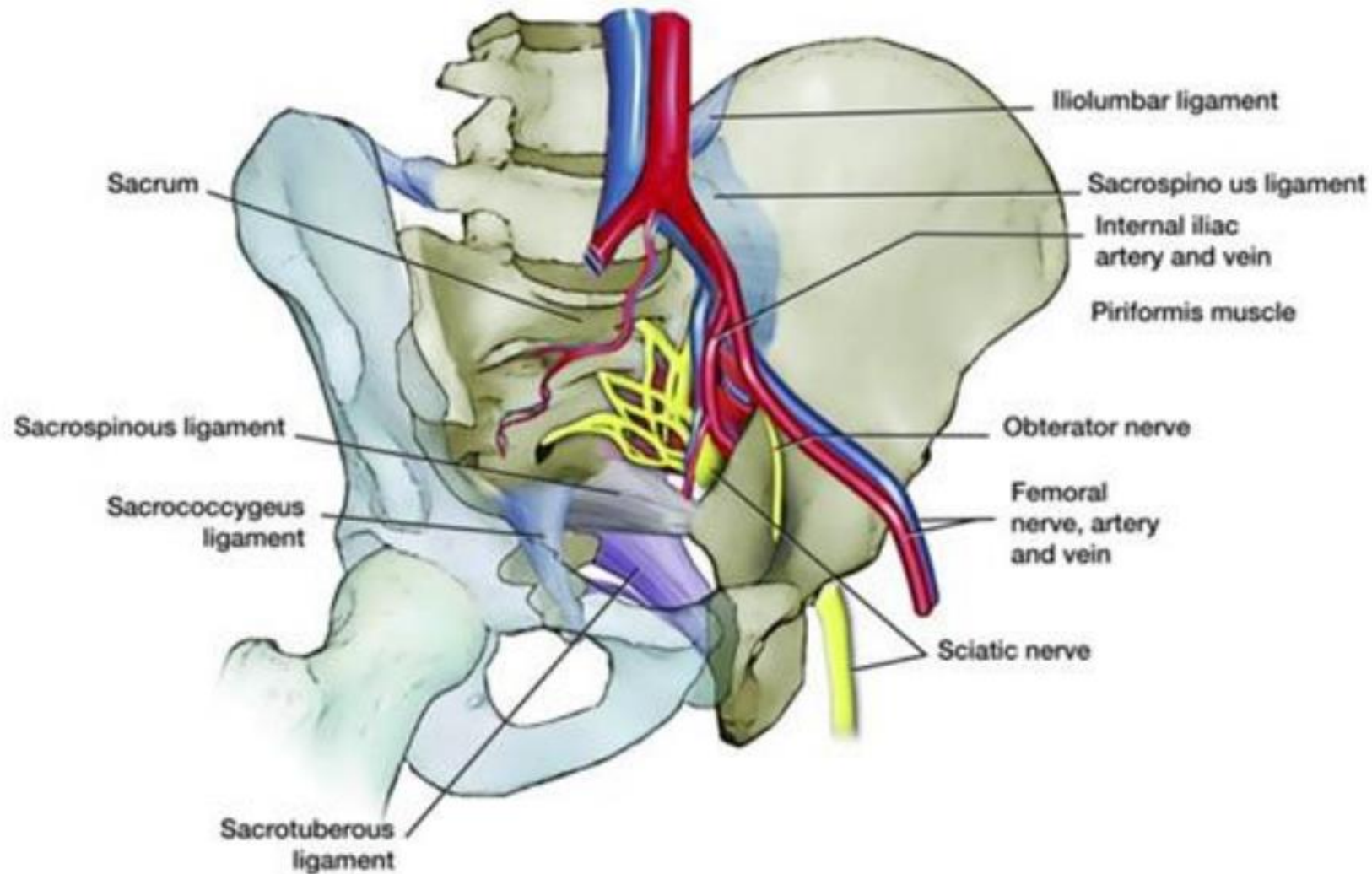


Pelvis and ligaments



Provide vertical and rotational stability

Neurovascular structure

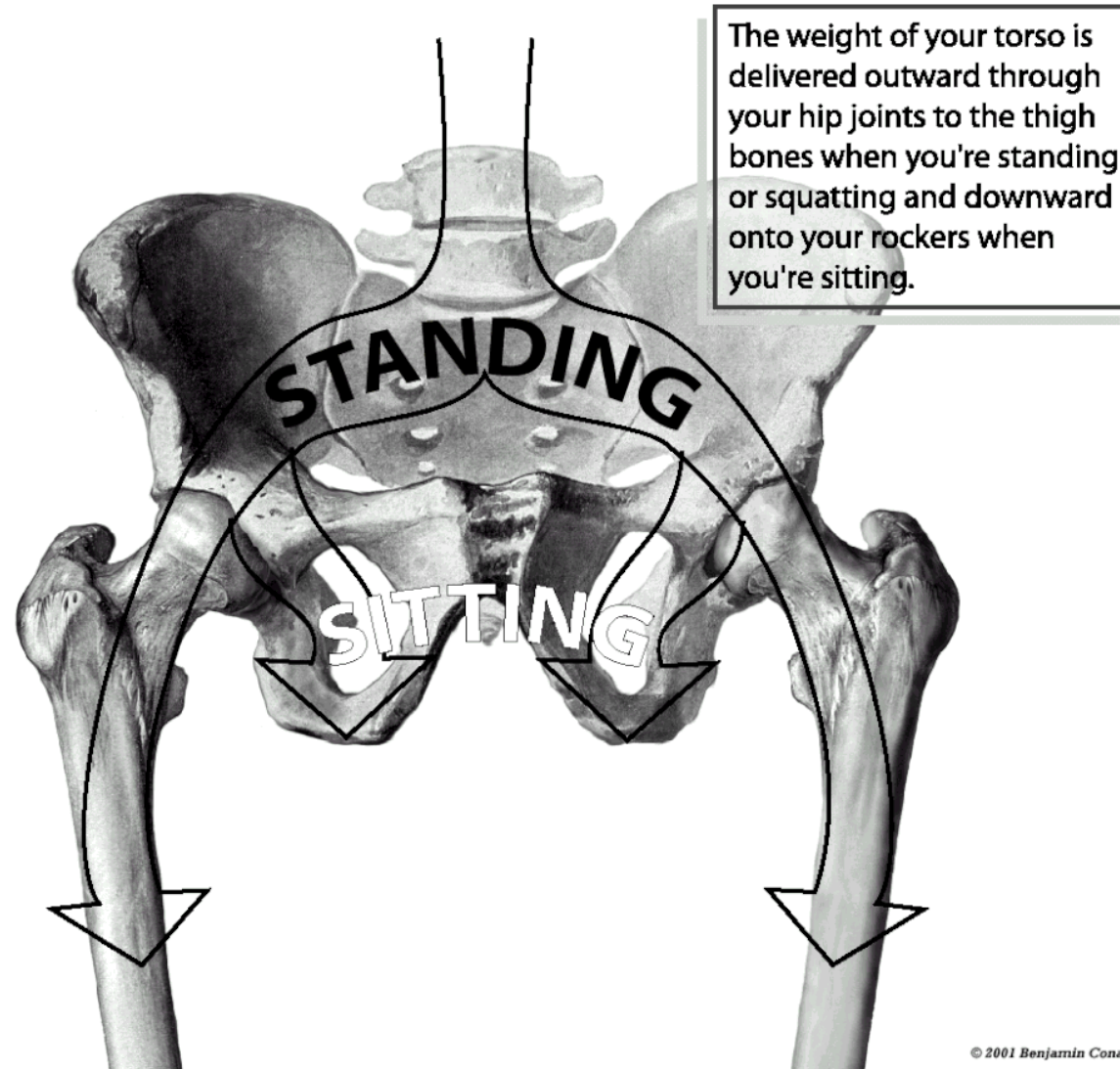




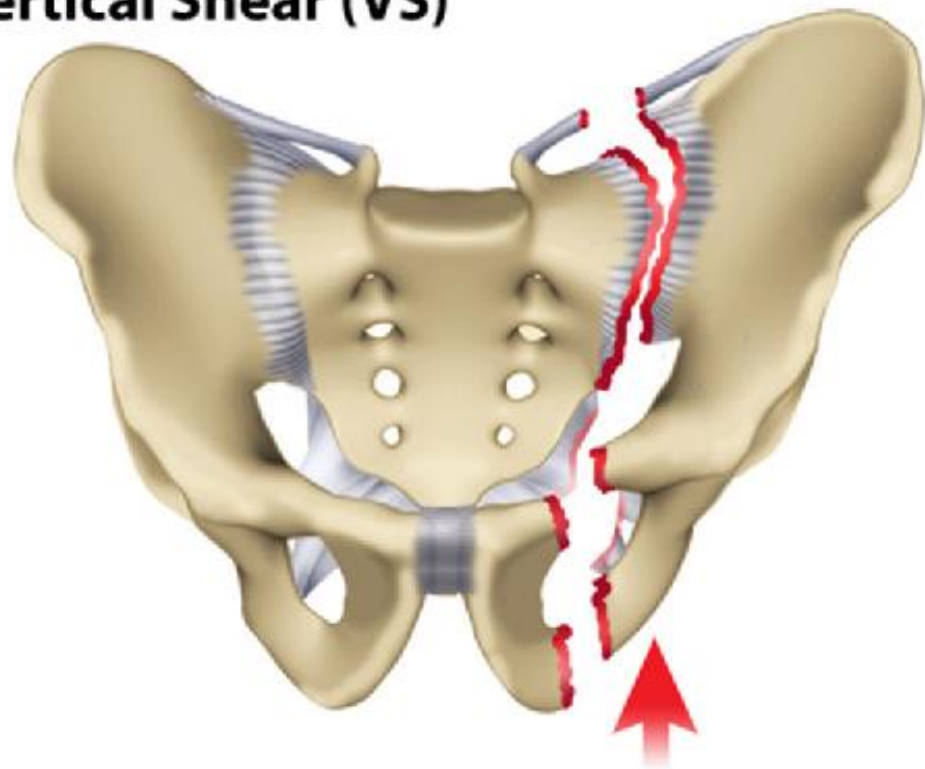
High associated injuries
Higher mortality rate

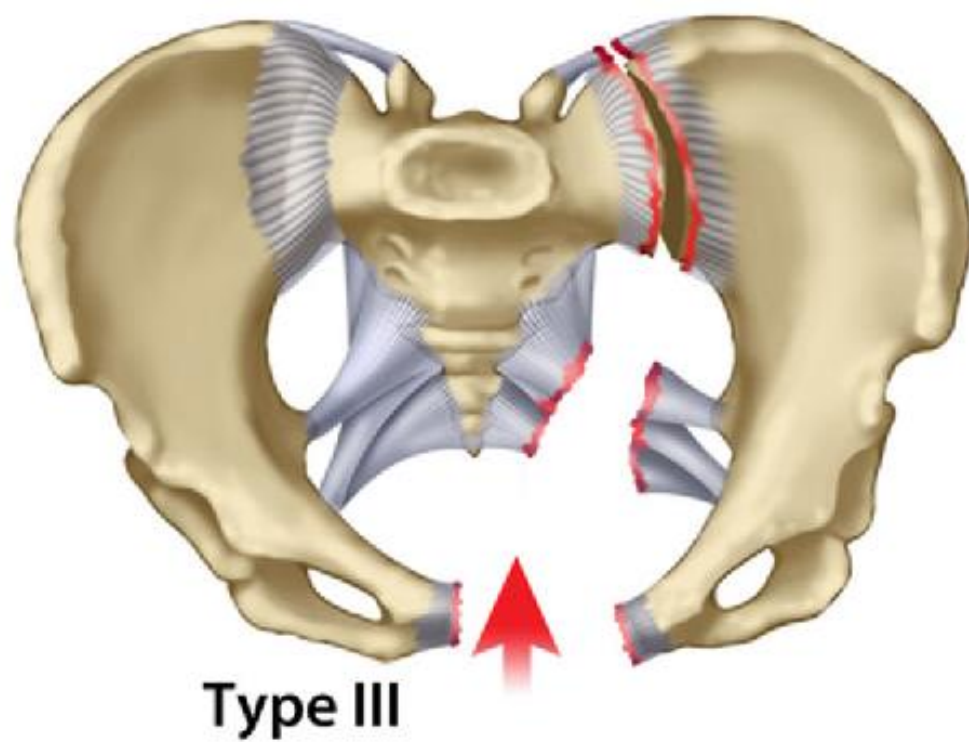


Biomechanics

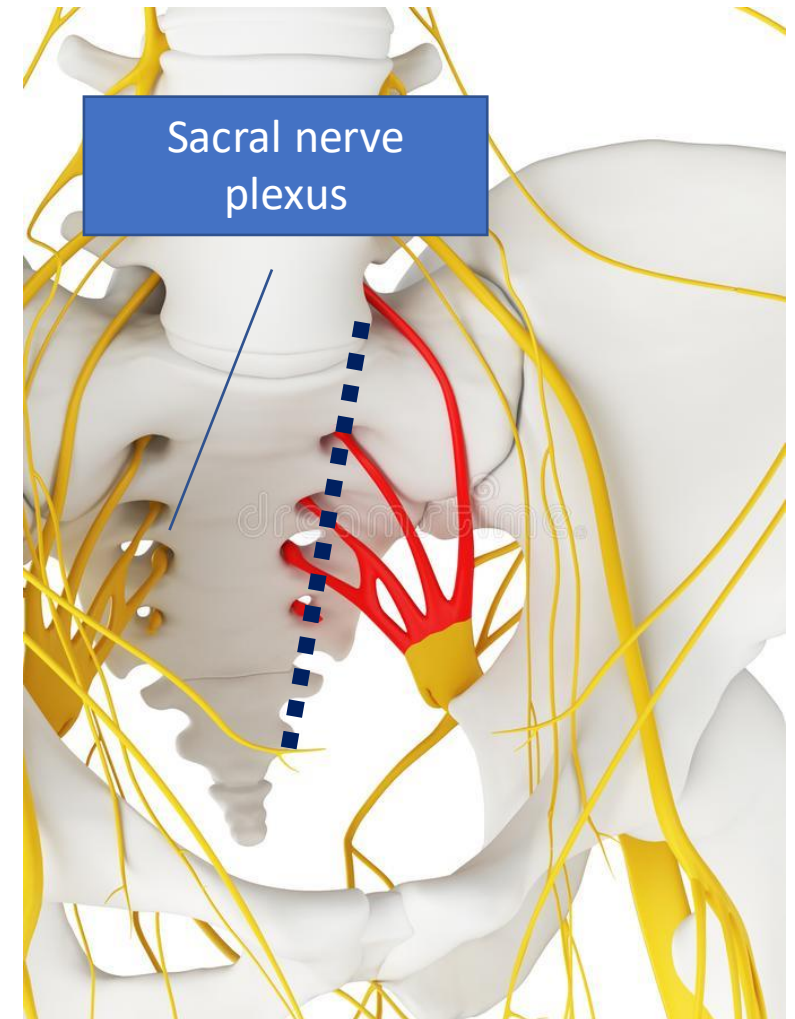
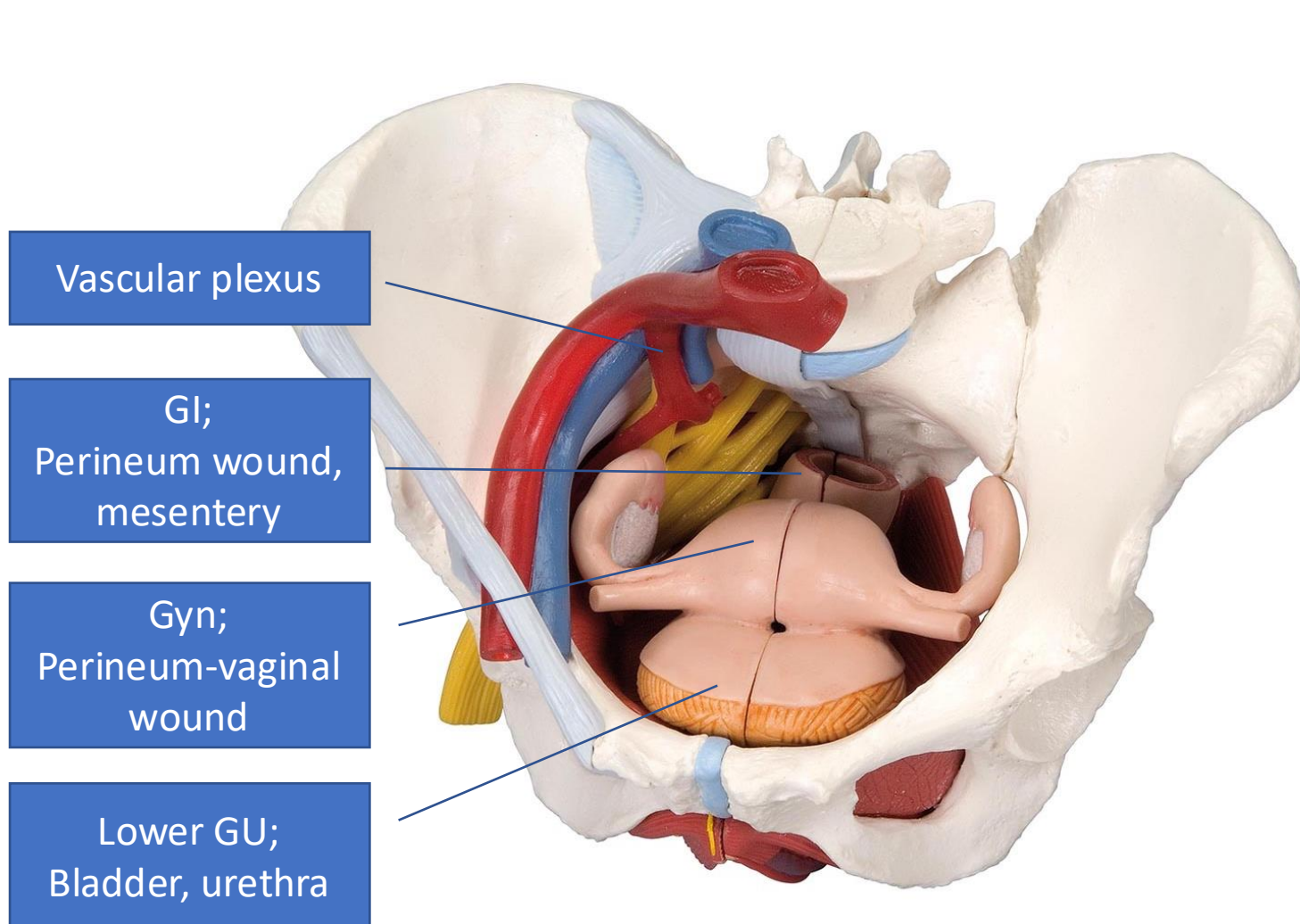


Vertical Shear (VS)





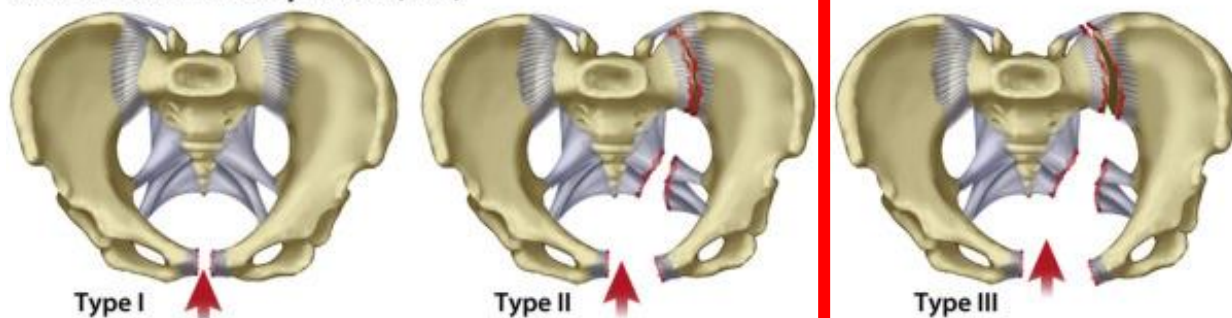
Associated injuries



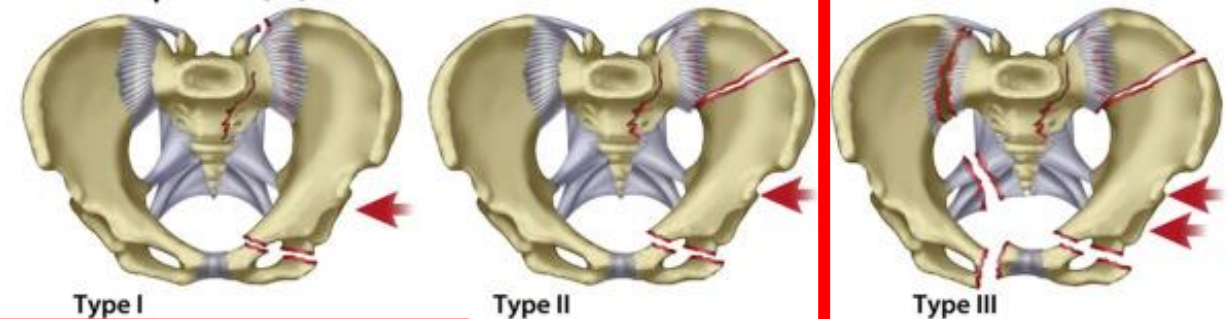
Associated injuries

Associated injuries	Incidence (%)
Closed-head injury	51
Long-bone fracture	48
Blunt abdominal injury (spleen, liver, GI tract, kidney, mesentery)	35
Peripheral nerve injury	26
Thoracic injury	20
Lower GU injury (bladder, urethra)	16

Anterior Posterior Compression (APC)



Lateral Compression (LC)



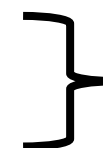
Vertical Shear (VS)



Young and Burgess's classification

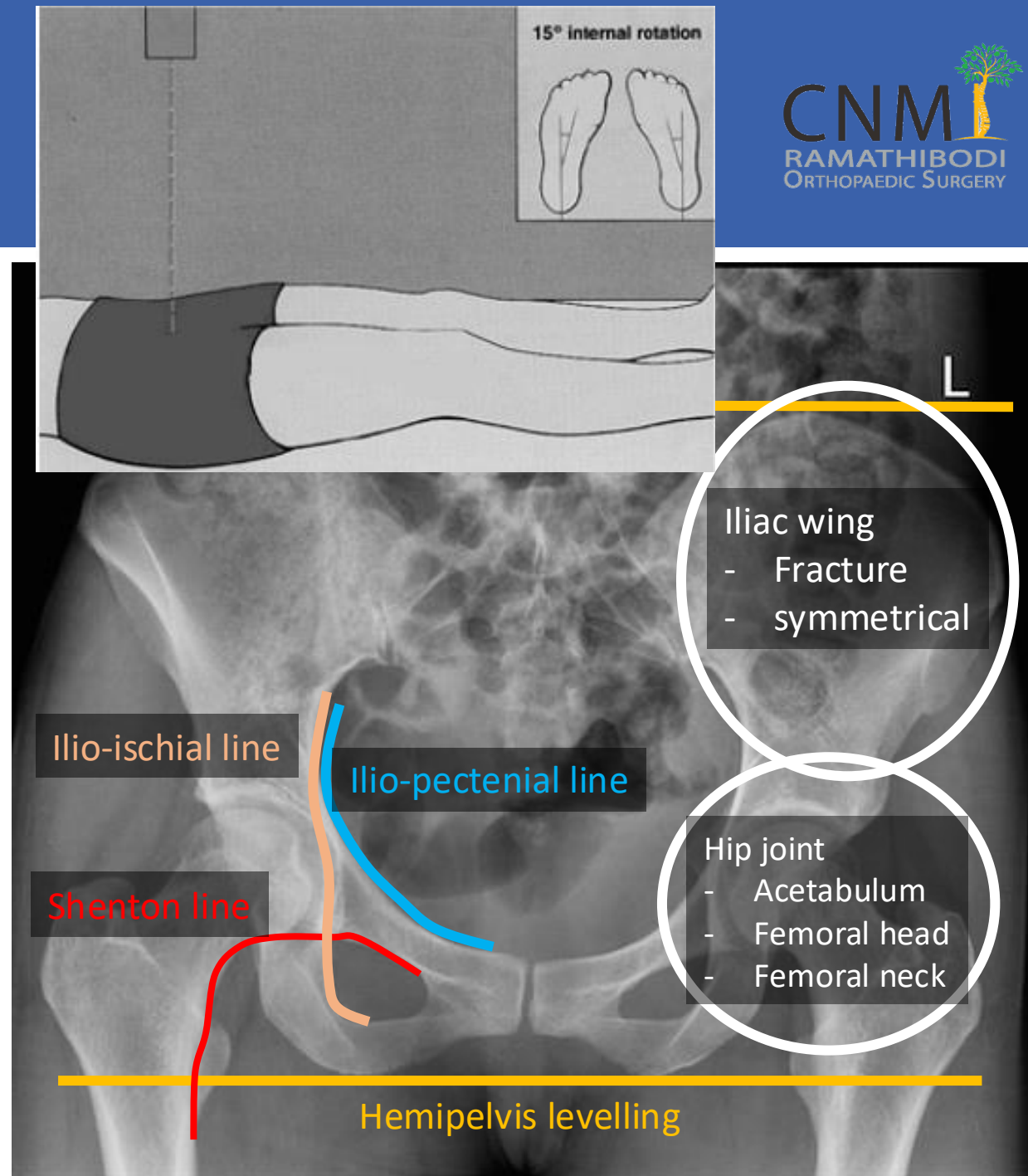
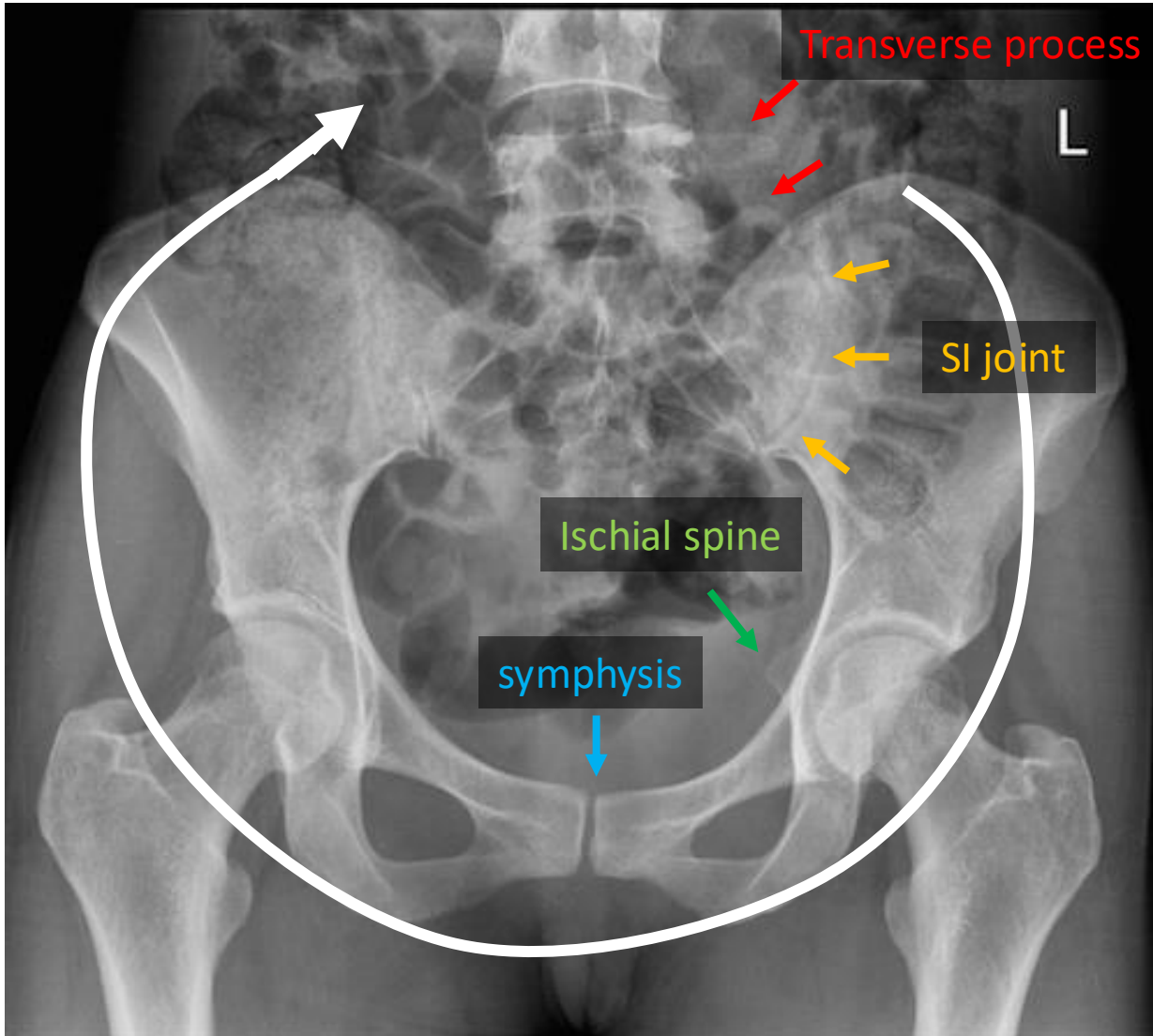
- Direction of forces reaction
- Degree of ligament disruption

Higher forces → Greater fracture displacement
→ Greater ligament disruption



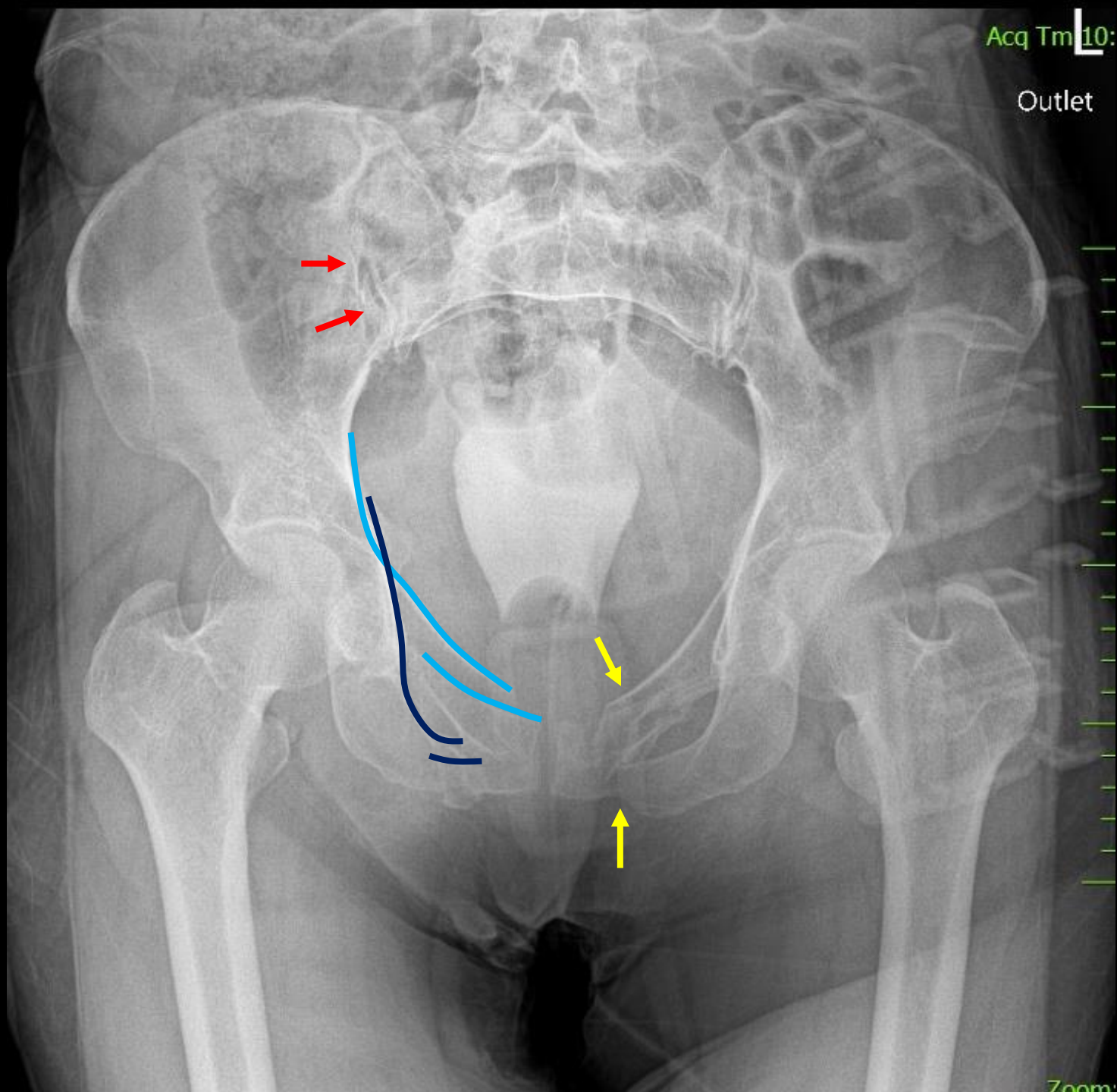
Higher associated injury risk

Plain radiograph



Acq Tm 10:

Outlet



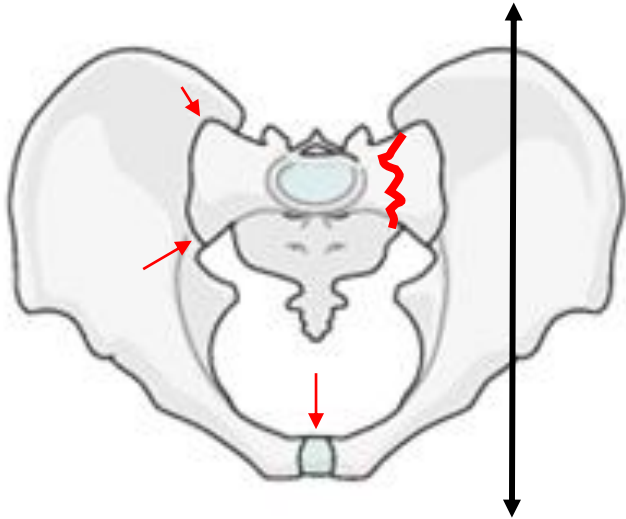
- 4 rami fracture = straddle fracture type
→ usually ass. with posterior ring fracture
- Rt. SI joint?

Need more imaging



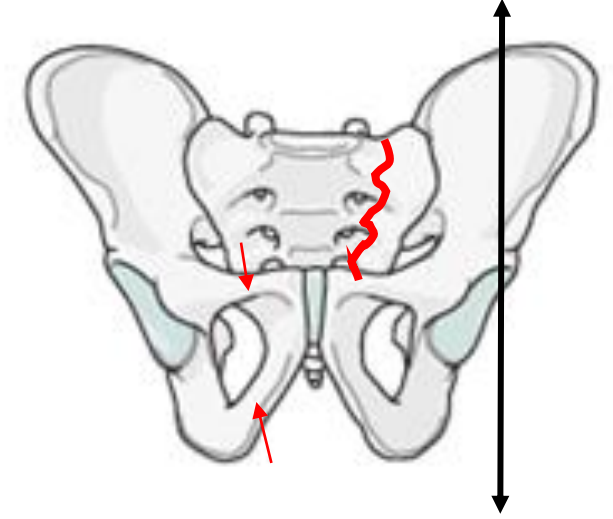
- Plain radiograph
 - pelvis inlet and outlet view
 - Judet's view (for acetabulum)
- CT pelvis with 3D reconstruction

Inlet view

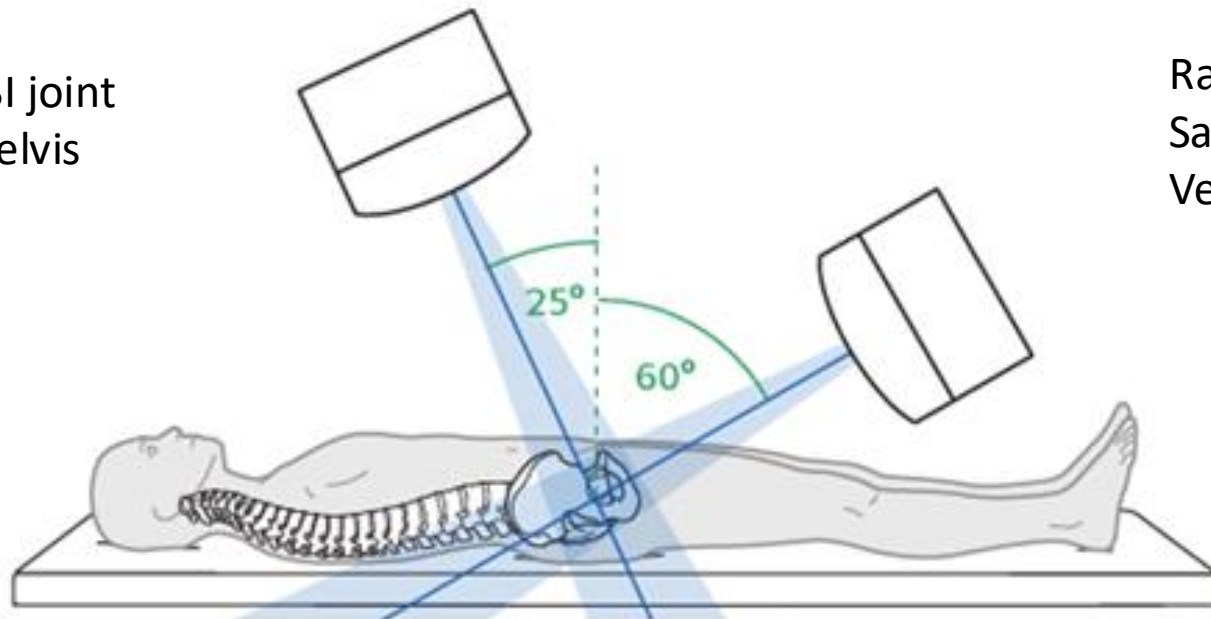


Anterior and posterior SI joint
AP translation of hemipelvis
Sacral fracture
Symphysis

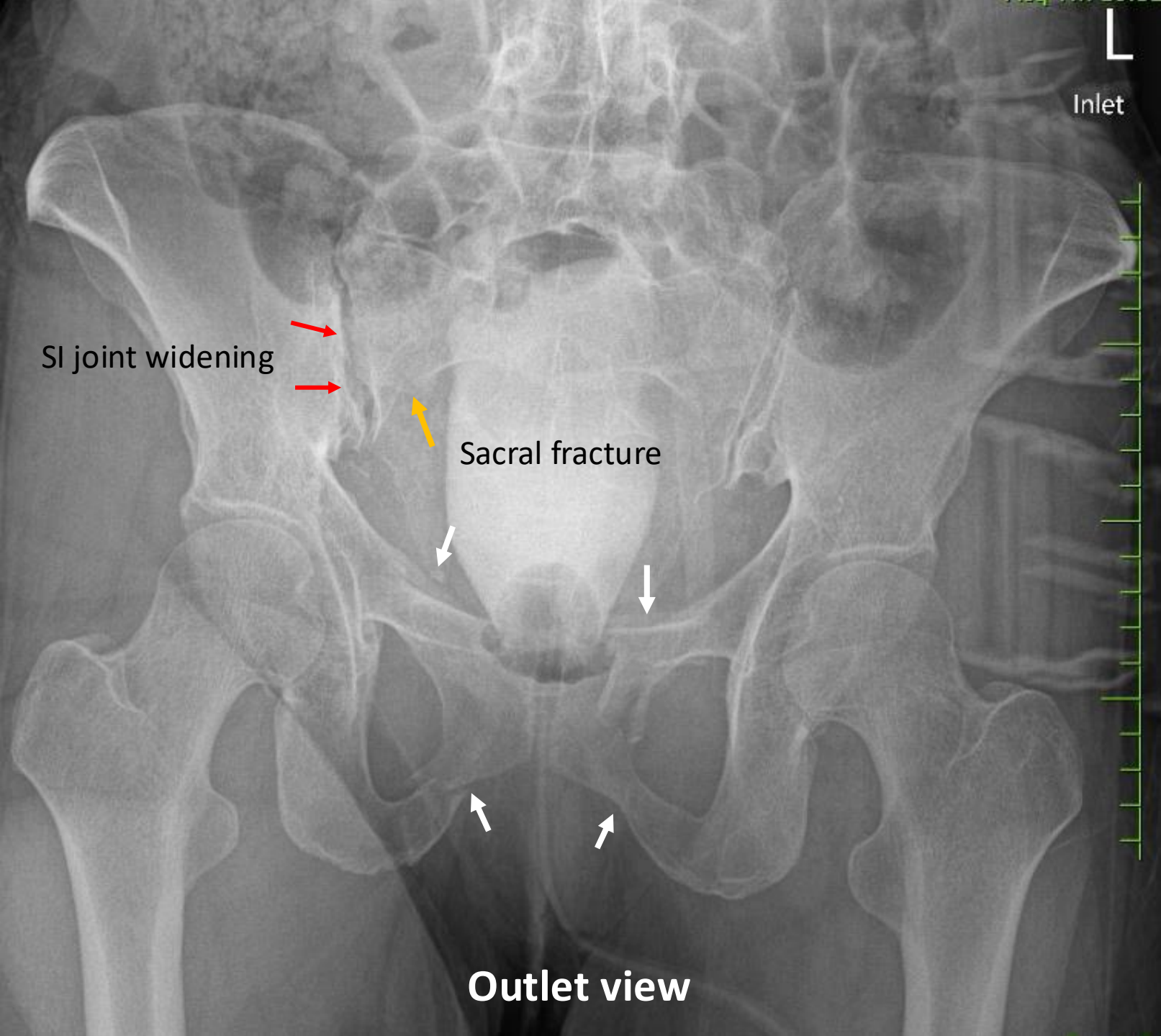
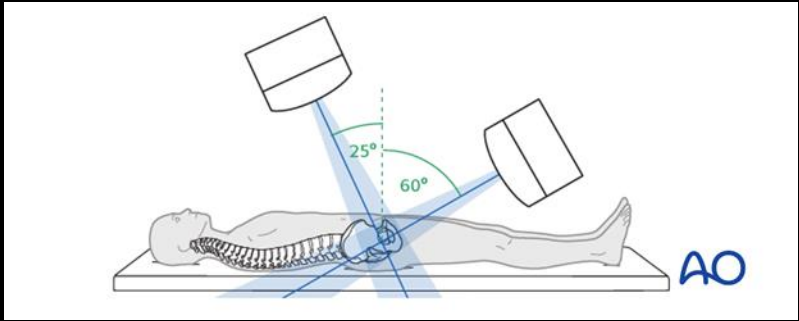
Outlet view

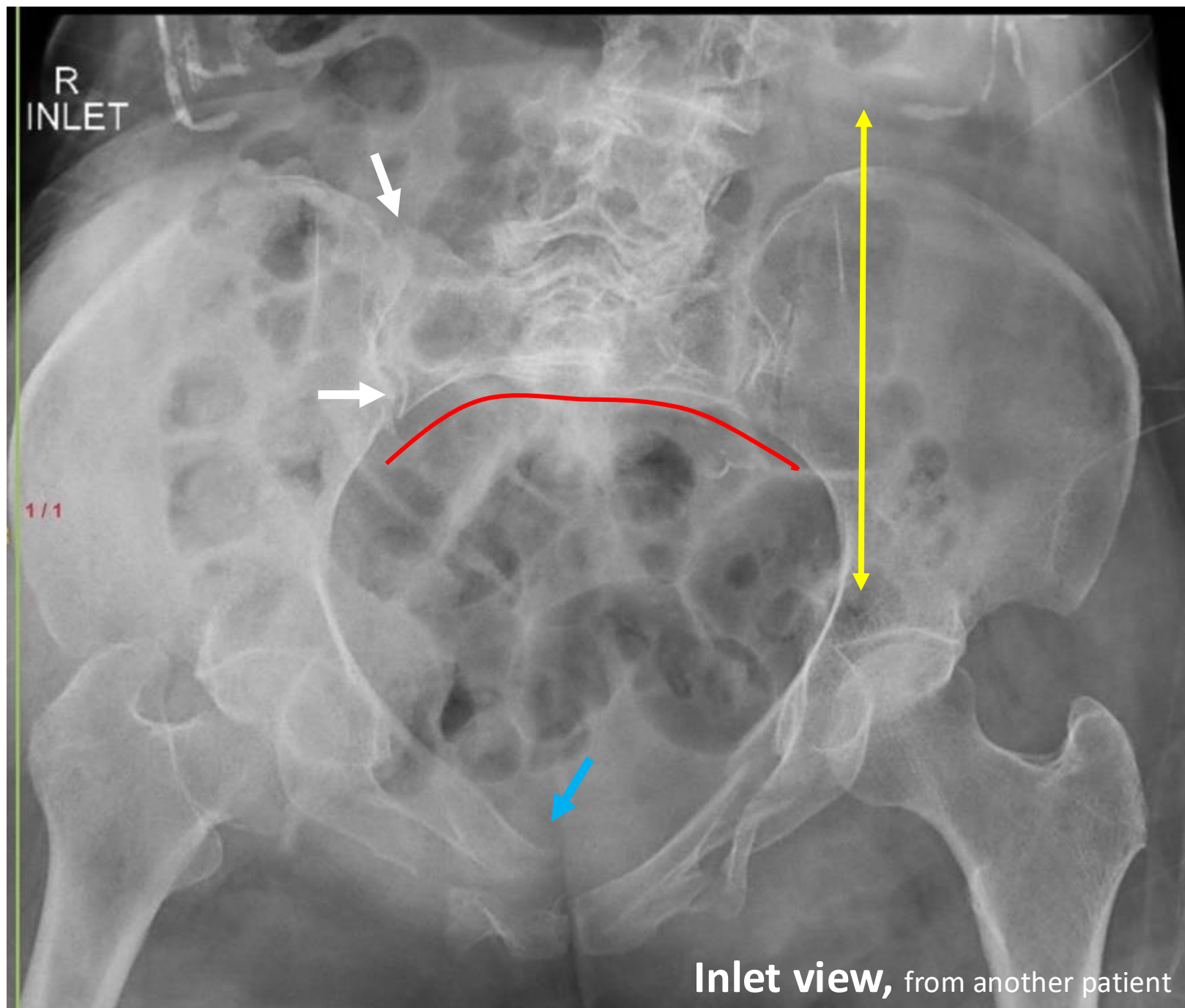
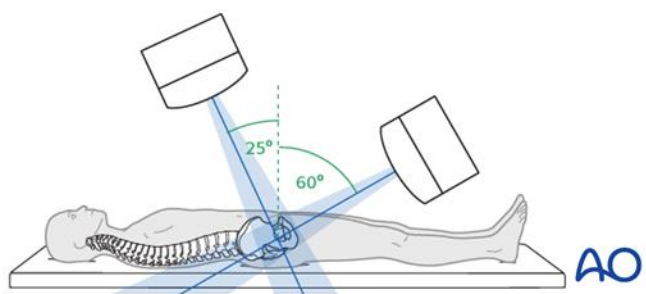


Rami fracture; superior & inferior
Sacral fracture
Vertical translation of hemipelvis

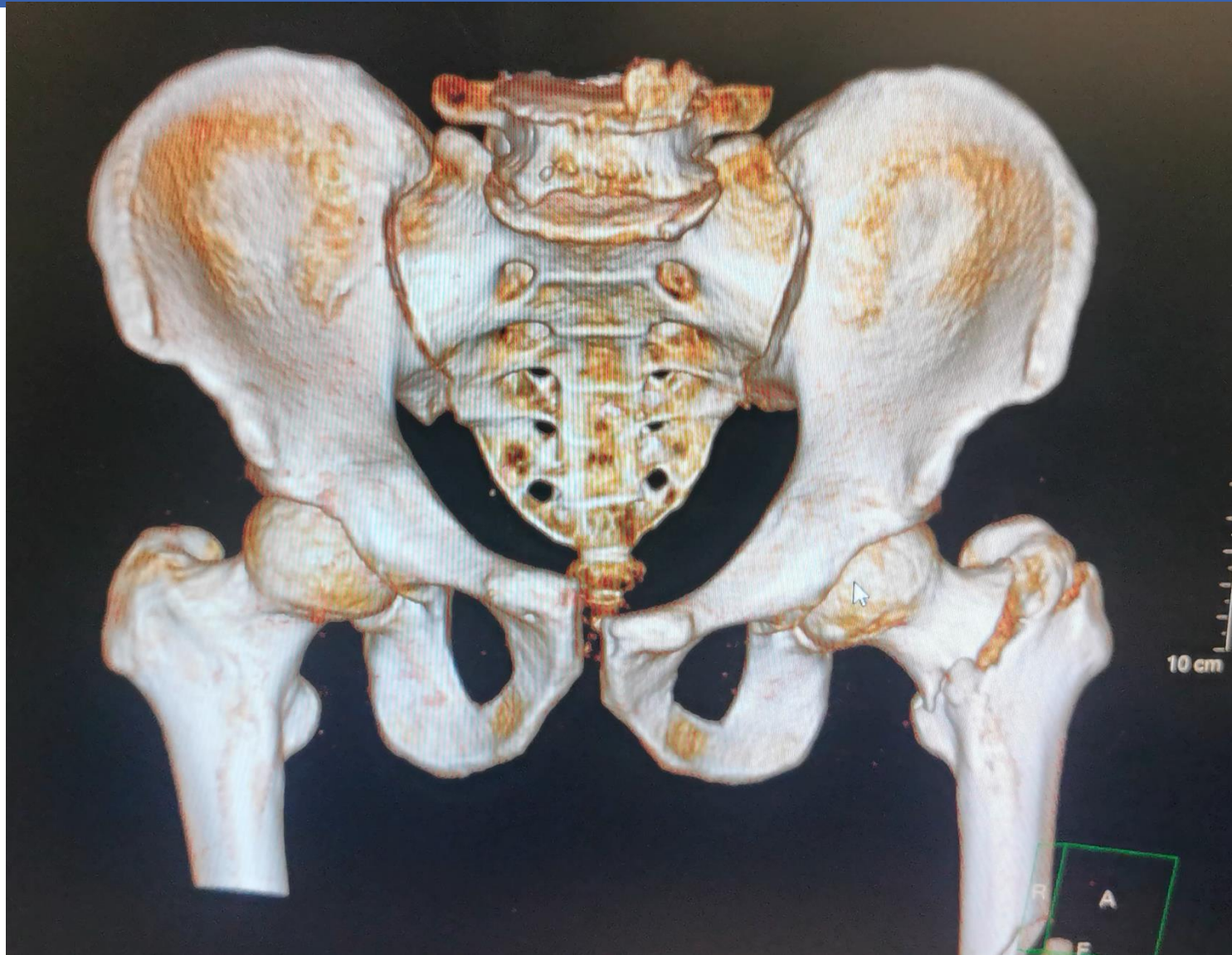


AO





CT pelvis with 3D reconstruction





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Patient assessment

Pelvis fracture

Patient assessment

Follow ATLS

- **Primary survey**

Airway, Breathing, Circulation, Disability,
Exposure

- Adjunct to primary survey and Resuscitation

hypovolemic shock

- **Secondary survey**

	Airway	Maintain airway with cervical spine control.
	Breathing	Assess breathing and ventilation. Apply high flow O ₂ .
	Circulation	Assess circulation with haemorrhage control.
	Disability	Check neurological status.
	Exposure/ Environment	Complete assessment of the patient but prevent hypothermia.

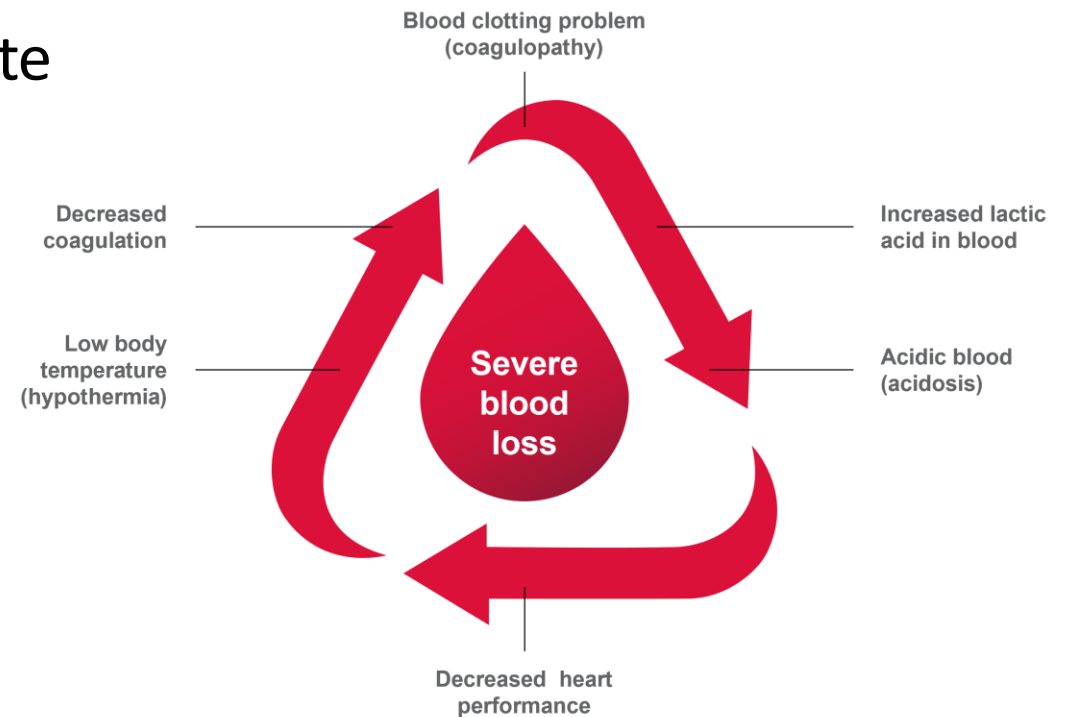
Classification of shock

Table 1: PHTLS & ATLS classifications of shock, seventh edition version

	Class I	Class II	Class III	Class IV
Blood loss (mL) (% total blood volume)	< 750 (< 15%)	750-1,500 (15-30%)	1,500-2,000 (30-40%)	> 2,000 (> 40%)
Heart rate (beats per minute)	Normal or minimally increased	> 100	> 120	> 140
Ventilatory rate (breaths per minute)	Normal	20-30	30-40	> 35
Systolic blood pressure (mm Hg)	Normal	Normal	Decreased	Greatly decreased
Urine output (mL per hour)	Normal	20-30	5-15	Minimal

Resuscitation

- **Large intravenous line**; at least 2 lines (16G or 18G)
- **Lab**; CBC, glucose, Electrolyte, BUN, Cr, lactate tube for matching blood component
- **Crystalloid solution**; loading 2 liters
- **Colloid solution**
- **Blood component**
 - Prepare for Massive blood transfusion (severely shock; LPRC: FFP: Plt. = 1:1:1)
 - 50-69% of unstable type pelvic fracture require at least 4 units of LPRC



Hemodynamically unstable

Causes of hypotension

- Blood loss; pelvis, blunt abdominal injury, hemothorax
- Intracranial cause; head injury- intracranial hemorrhage
- Neurological cause; spinal shock
- Pressure effect; tension hemothorax

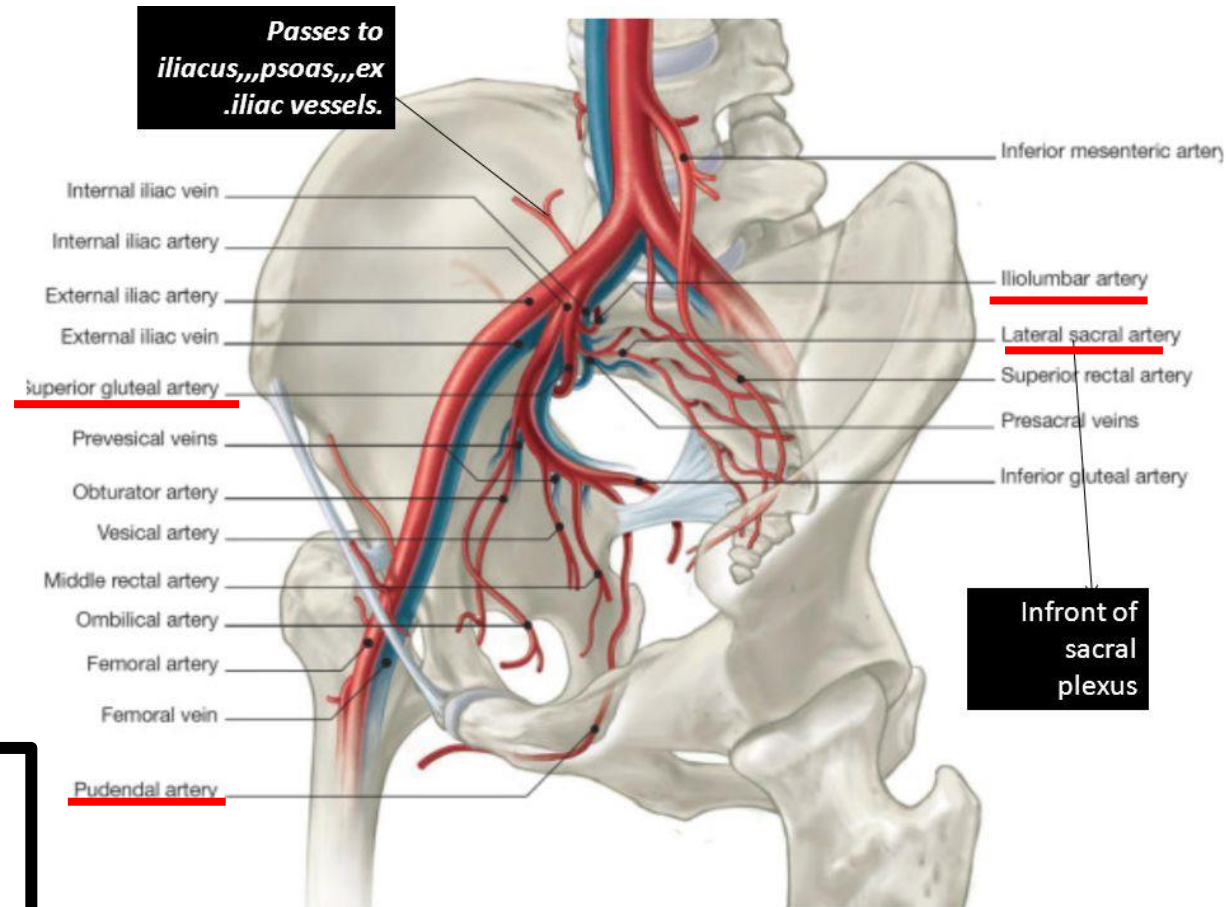
Physical examination during primary survey
FAST ultrasound
CT; brain, whole abd, pelvis

Hypovolemic shock

- From associated injury (35-40%) ; blunt abdominal injury, chest injury
- From pelvic fracture
 - Osseous bleeding
 - Vascular bleeding

Iliolumbar plexus
Lateral sacral artery
Internal iliac artery
Internal pudendal plexus
Sacral venous plexus
Superior gluteal artery (most common artery injury)

80% bleeding from venous plexus
20% from arterial source



Secondary survey

- **“AMPLE”**

Allergies

Medications

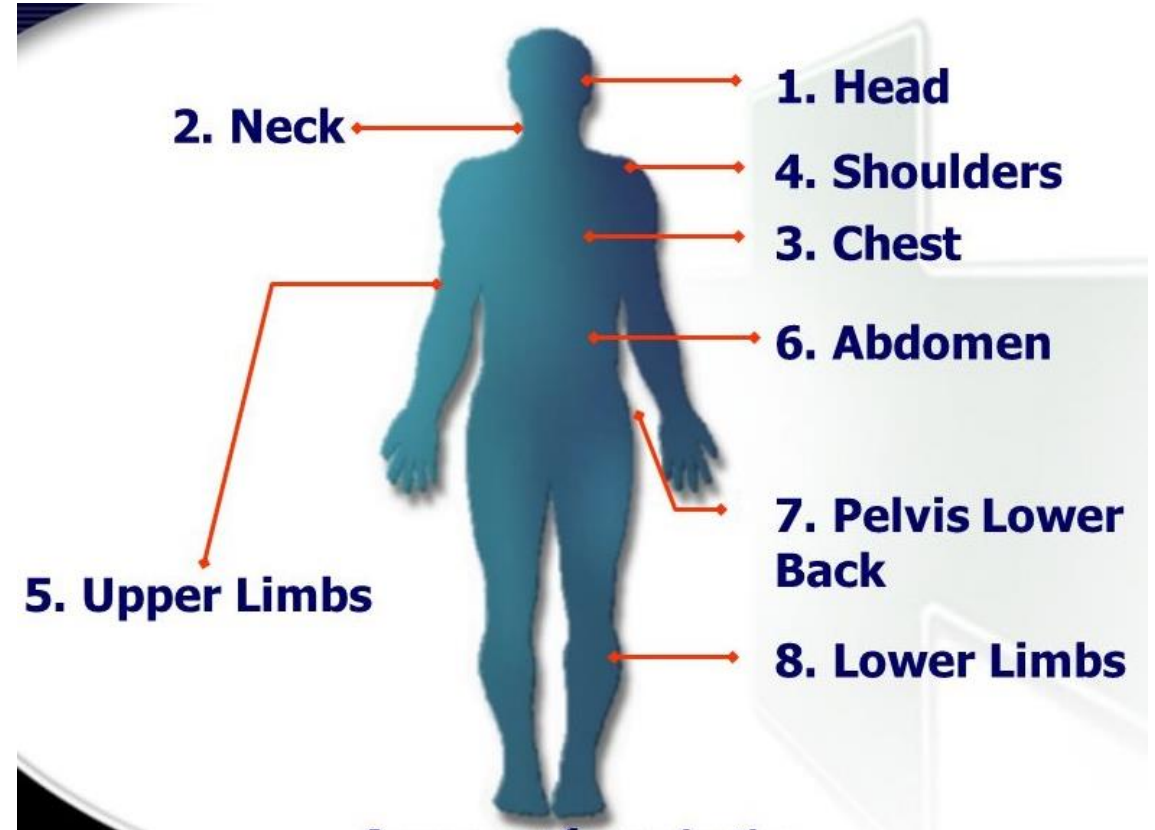
Past illnesses

Last meal (time)

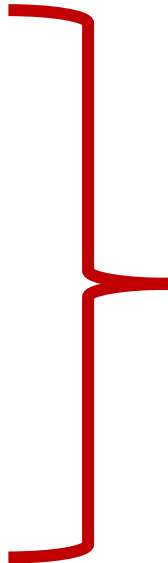
Events (preceding injury)

- Clarify mechanism of injury
- Assess for other condition

Physical examination from head to toe, hypoglycemia, toxin



Pelvic fracture examination

- **Bone test**: pelvic stability test → determining fracture and stability
 - **Rectal examination** (PR)
 - Rectal and perineal injury
 - Urethral injury; high-riding prostate (men)
 - Perianal sensation, anal constriction
 - **Per vaginal examination**
 - Vaginal bleeding
 - **Urethral examination**; bleeding per urethra
 - **Neurological examination**; sacral plexus, sciatic nerve condition (L5,S1)
 - **Soft tissue injury**; Moral-Lavalle lesion
- 
- Open pelvic fracture**

Pelvic compression test

- Check stability of pelvis (**DON'T REPEAT**)
 - 1) Apply gentle medial pressure with palms by pressing inward on iliac crests
 - 2) With patient supine, apply gentle posterior pressure by pressing downward on iliac crests
 - 3) Apply gentle downward pressure on pubis to check pelvic ring stability



Medial pressure (1)



Posterior iliac pressure (2)



Posterior pubis pressure (3)

Open pelvic fracture

- Open fracture = fracture + wound (that exposed the fracture)
- High risk for bone infection (osteomyelitis) → difficult to treat

External wound



Internal wound



- Immediate iv antibiotic: Broad spectrum cefazolin 1 g + gentamicin 240 mg + penicillin 1.2 mu
- Tetanus prophylaxis
- Urgency surgical debridement

Soft tissue injury; Morel-lavalle lesion



Day 1



Day 3

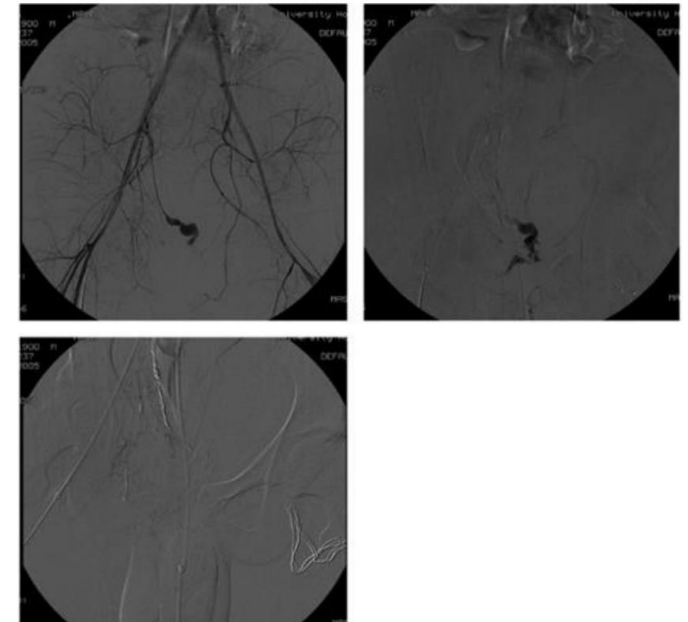
- Closed degloving injury
- Results from shearing force → subcutaneous tissue torn → disruption of blood supply to overriding skin
- Hematoma, liquefied fat, skin necrosis
- 1/3 become infected
- Need surgical debridement



Day 7

Special intervention

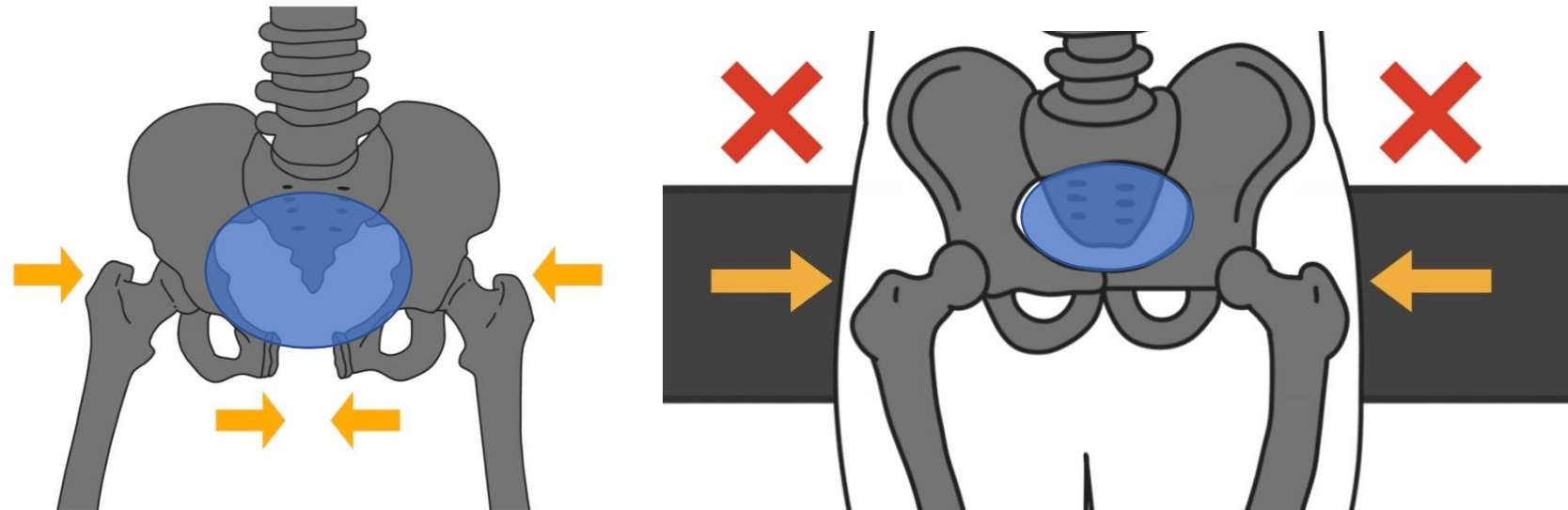
- Angiogram and embolization
- Stop arterial bleeding; superior gluteal artery
- Disadvantage;
 - Source of arterial bleeding is identified in only 10-15%
 - Does not address venous bleeding



pelvic stabilization

Principle

- Immobilization → Reduce fracture motion
→ Decrease pain
→ facilitate blood clot formation
- Reduce pelvic volume → increase intra-pelvic pressure effect
→ reduce blood loss



Pelvic stabilization



Pelvic sheet ✓

- Initially, the application is **recommended for all cases** with suspected pelvic fractures.
- Remove within 24 hours



Pelvic binder ✓



Anti-shock garment

Pelvic binder application

Remove as soon as possible or within 24 hours

Pelvic Binder Application

Do Not Place over the
Iliac Crest!

Place over the
Greater Trochanters

@aLittleMedic

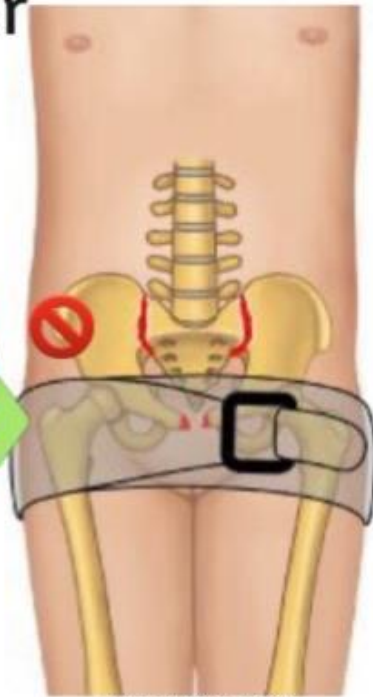
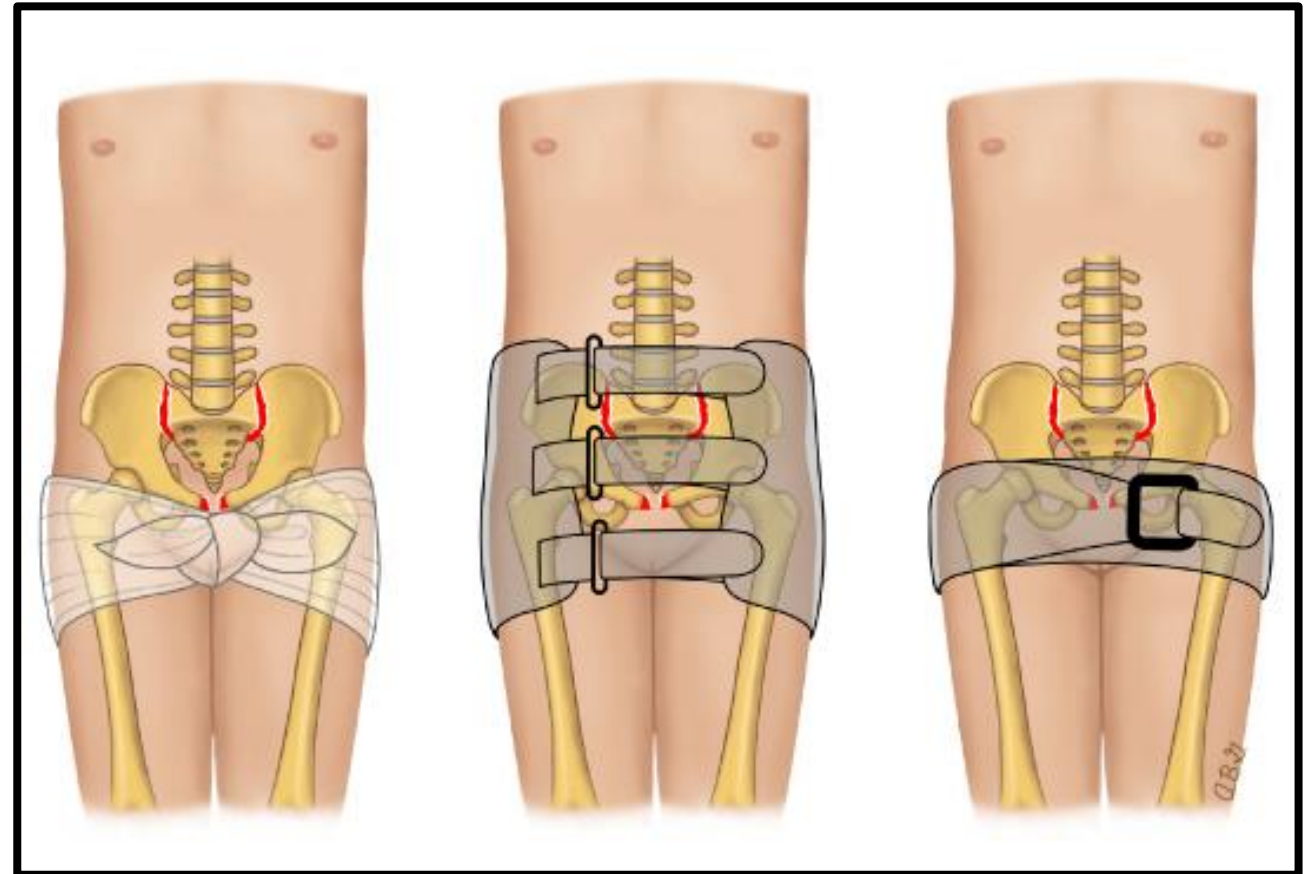


Image: @PHEM_cast

Loose or Extremely Tight are Both Bad



Pelvic binder



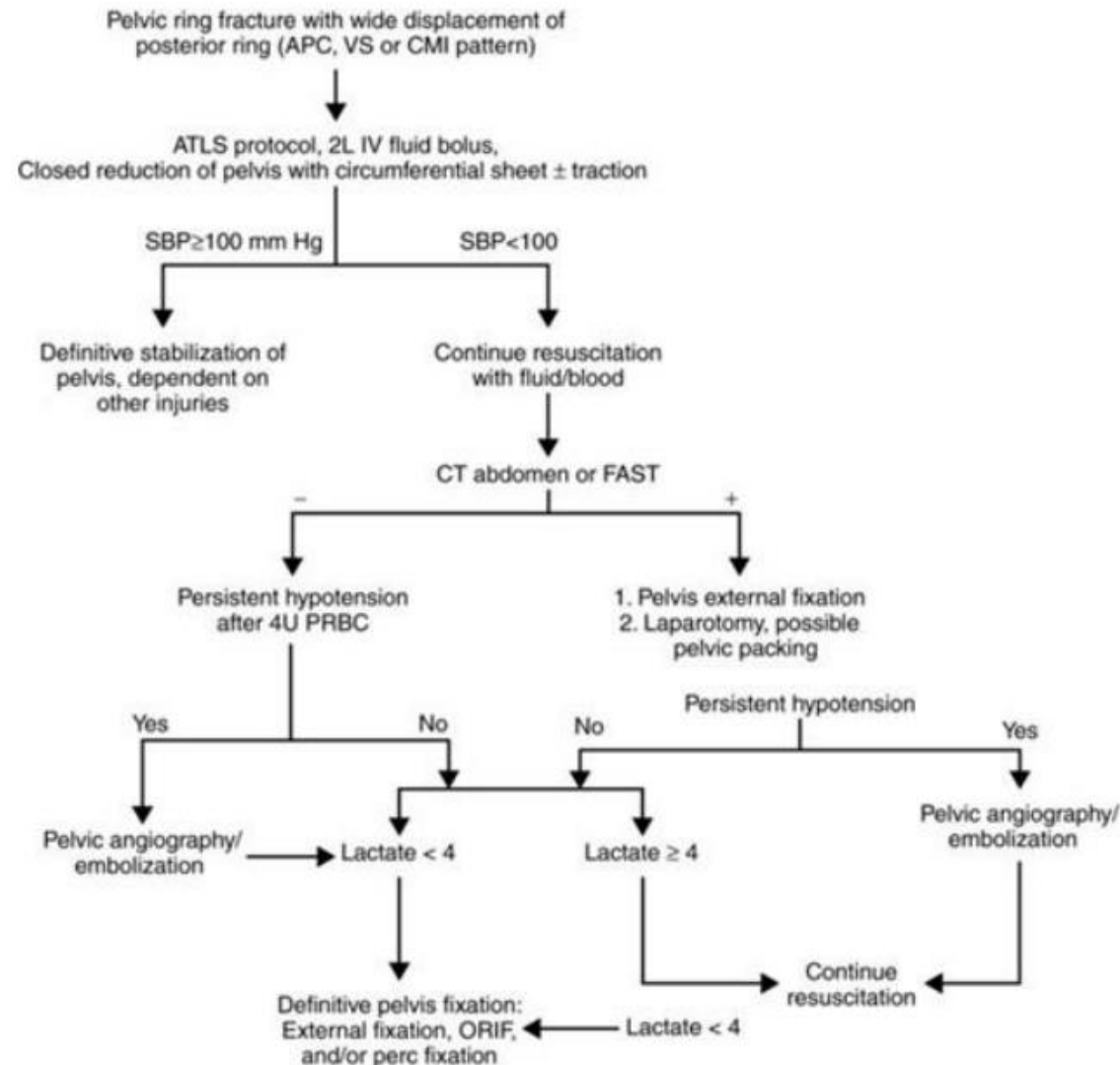
Pelvic sheet



Pelvic sheet



Management step conclusion





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Case



Male 22 years old; car accident

Primary survey ABCDE

Good consciousness

Vital sign;

Temp 37.3, BP 80/50, PR 113, RR16
O2sat 98%



Immediate management primary survey

Fluid resuscitation

0.9%NSS loading 2 Liters, 2 large lines

Blood component group matching

Lab examination; Hct, CBC, Electrolyte,

BUN,Cr, Coagulogram, Glucose and lactate



Do it simultaneously

Secondary survey

AMPLE & head to toe examination

Severe pain left hip, can not move his Lt leg

No open wound

Otherwise normal

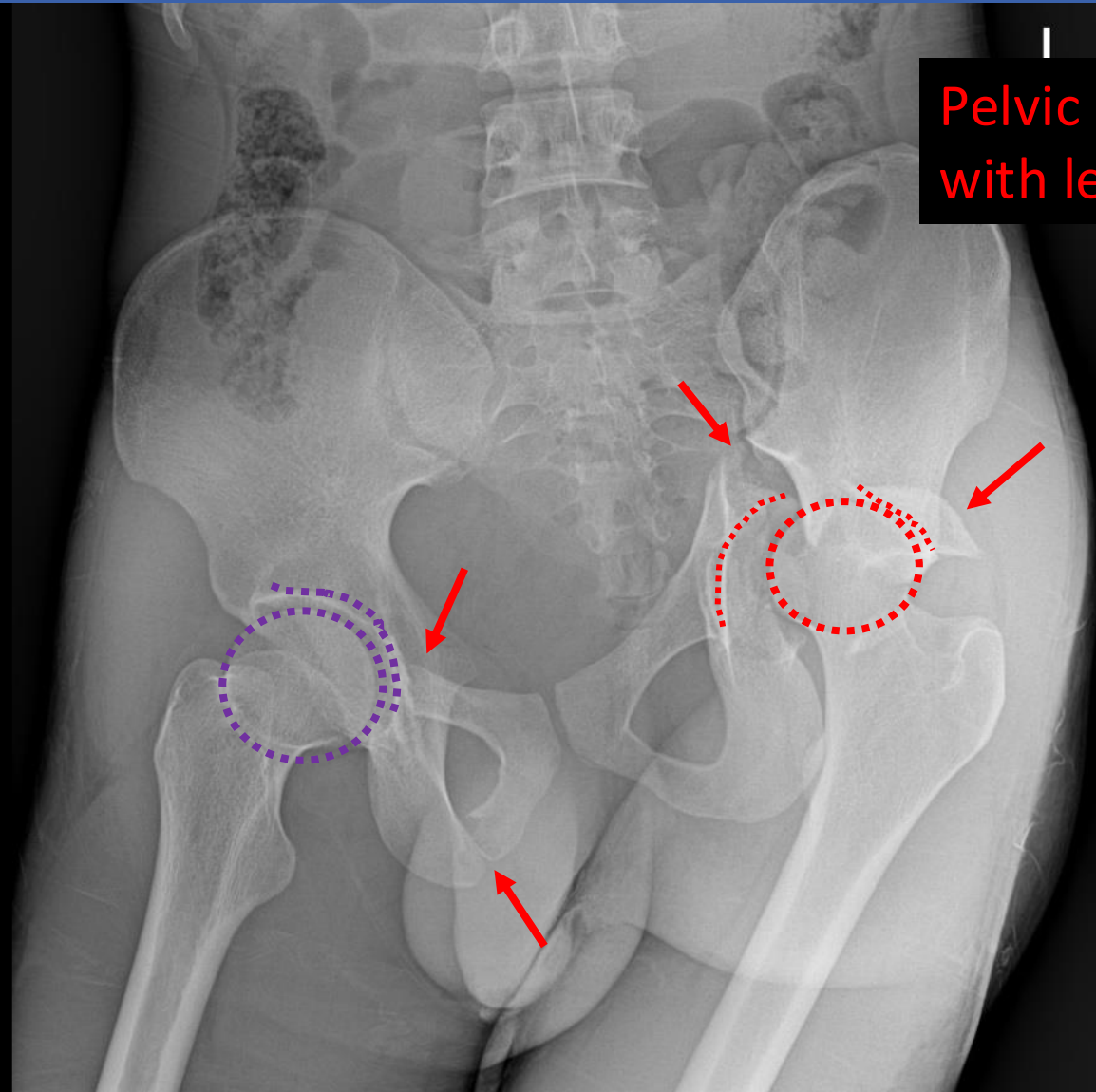
Consult general surgery and orthopedist



Procedure; Pelvic binder

Investigation Plain radiograph;
C-spine lateral, Chest supine AP,
pelvis AP film

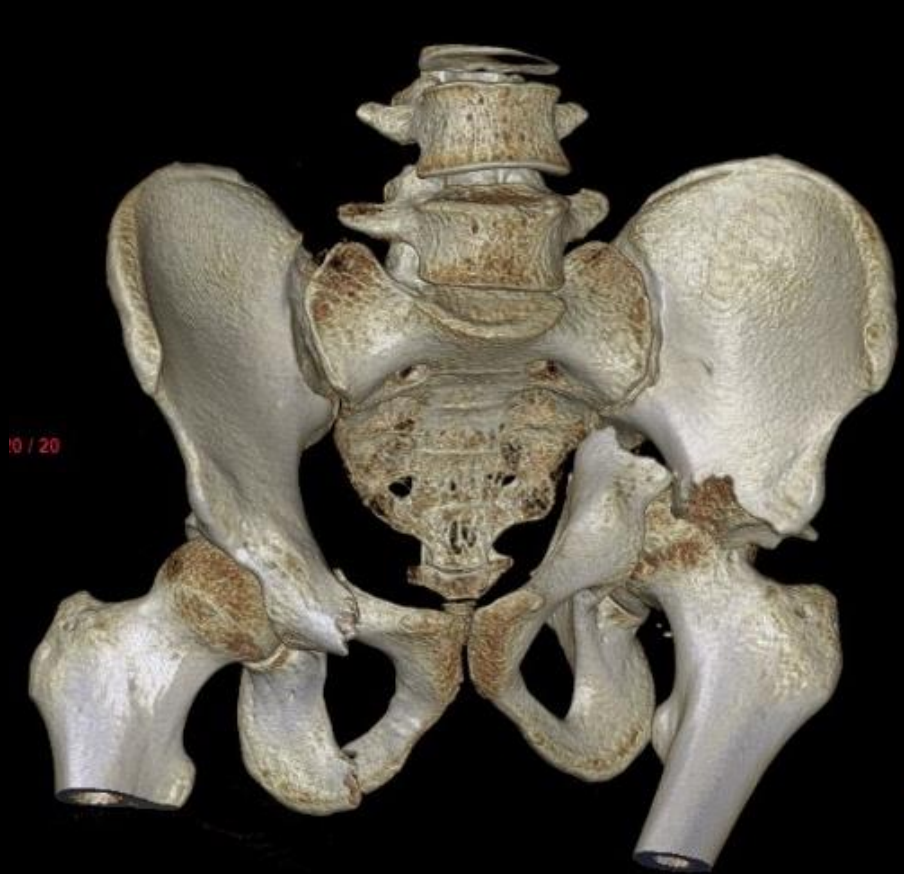
Male 22 years old; car accident

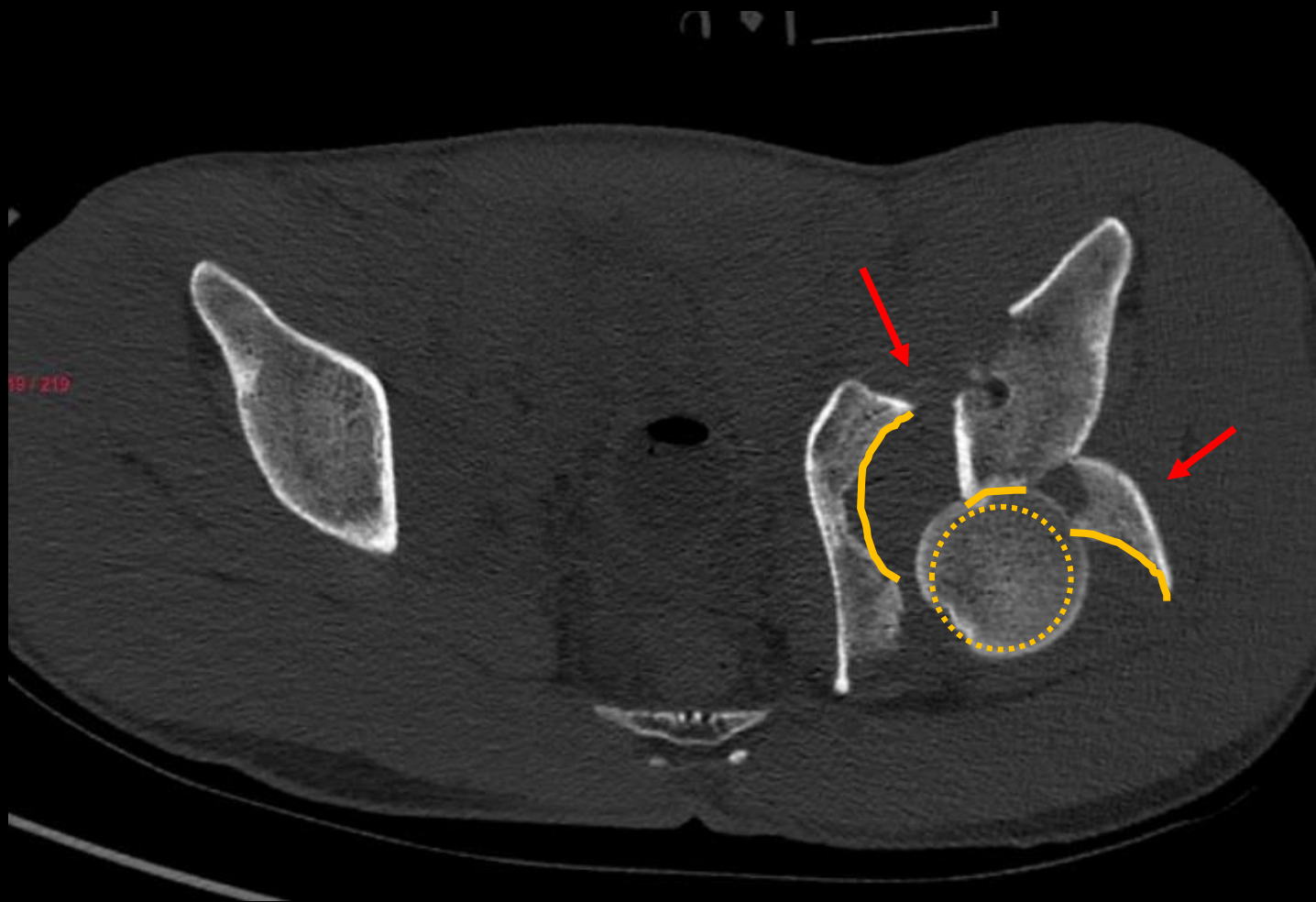


Pelvic and left acetabulum fracture
with left posterior hip dislocation



CT scan at pelvis with
3D reconstruction





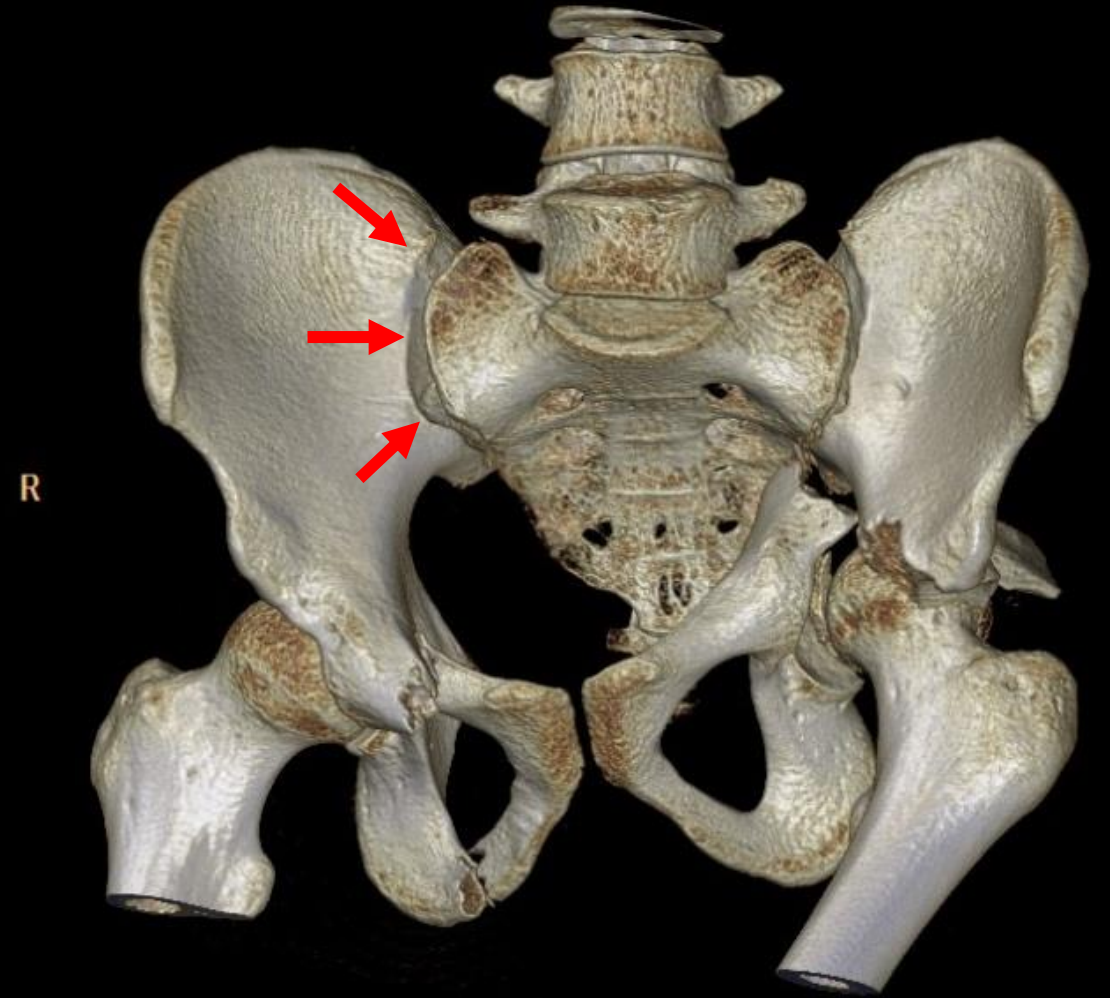
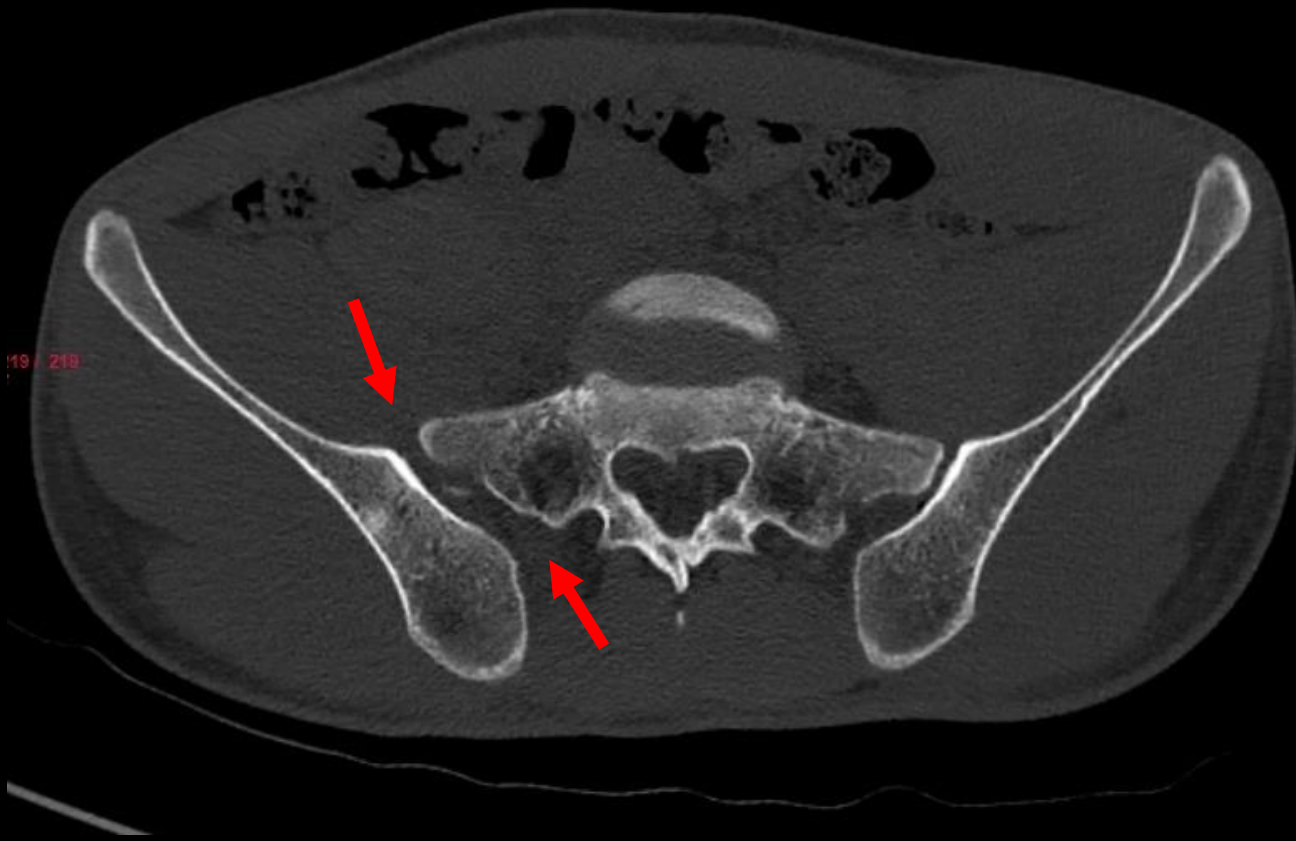
Acetabulum fracture with
posterior hip dislocation



Orthopedic urgency condition

Pelvic fracture; LC type

Left acetabulum fracture with posterior hip dislocation

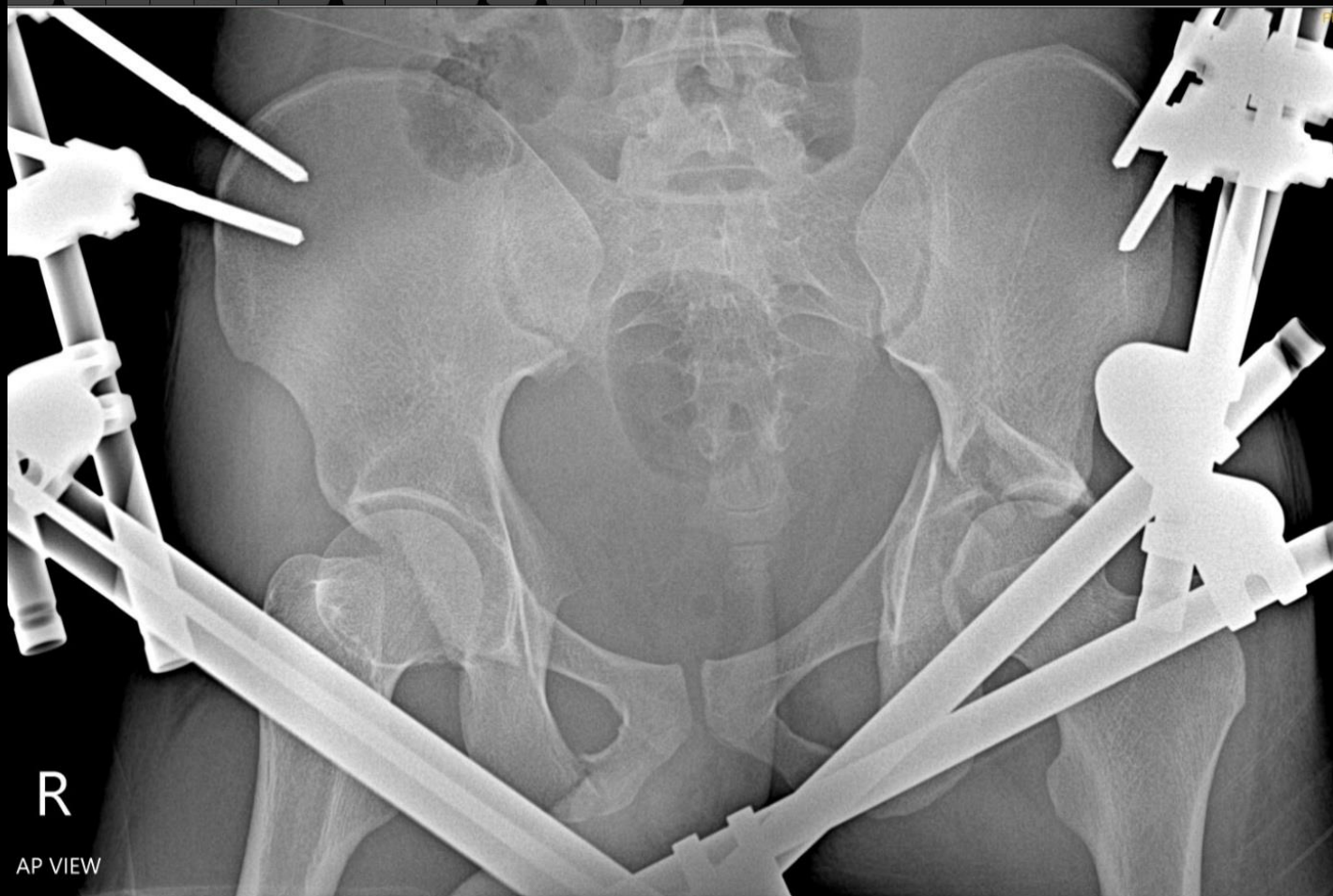


Sacroiliac joint separation



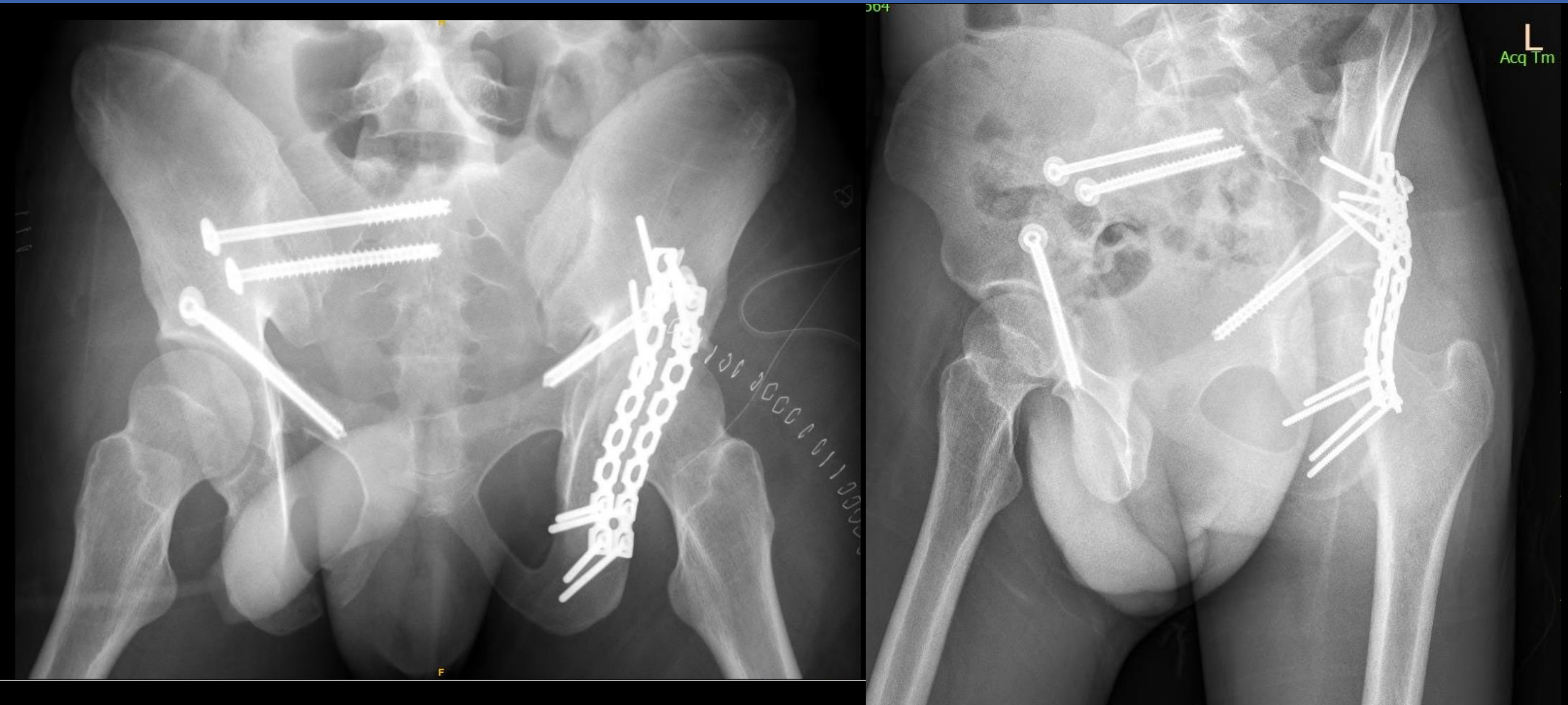
Unstable pelvic fracture; LC type

Management



- Set OR under general anesthesia
1. Closed reduction left hip
 2. Pelvic external fixation

Definitive management



Male 15 years old; MCA

- Motorcycle crush by a car
- His consciousness was temporally loss and regain conscious at emergency room.

Primary survey



Primary survey ABCDE

- Spontaneous breathing without stridor
Chest compression test; negative
- Fully actively neck flexion
- Vital sign; BP 70/40 mmHg, PR 121, RR 16, Temp 37.5
- Can not move right & left leg by pain, abdominal pain
Back not pain

Male 15 years old; MCA

Primary survey ABCDE

- Spontaneous breathing without stridor
Chest compression test; negative
- Fully active neck flexion
- Vital sign; BP 70/40 mmHg, PR 121, RR 16, Temp 37.5
- Can not move right & left leg by pain, abdominal pain
Back not pain



Fluid resuscitation

- 2 large IV line (16 or 18G)
- 0.9% NSS 1000 ml loading or pushing x2 (2 liters)
- Group matching; LPRC 4 units
- Blood exam; glucose, lactate, Hct, CBC, Electrolyte, Coag, BUN, Cr

Secondary survey

AMPLE & head to toe examination

- Abdominal guarding and pain
- Bleeding per urethra
- Abrasion Rt flank with laceration wound
- Pelvic compression test; pain



Male 15 years old; MCA

Secondary survey

AMPLE & head to toe examination

- Abdominal guarding and pain
- Bleeding per urethra
- Abrasion Rt. flank with laceration wound
- Pelvic compression test; pain

Consult general surgery, orthopedist, uro-surgery nephrosis.

Pelvic binder

Ultrasound FAST → positive at suprapubic area

Plain film; C-spine, chest supine AP, pelvis AP

CT whole abdomen (if available) – include pelvis

****DO NOT INSERT FOLEY CATH ****

CT WHOLE ABDOMEN INCLUDED PELVIS: PRE and POST contrast enhancement.

Hx : Blunt abdominal and pelvic injury

Findings :

- The study reveals moderate amount of free fluid throughout the lower abdominal and in pelvic cavity. Evidence of extraluminal contrast/ small free gas into perivesicle space and extravasation of urinary tract contrast media below the urogenital diaphragm and periurethral region and after delayed scan which increased extension of extraluminal contrast and abnormal position of foley's balloon catheter, suspicious of extraperitoneal rupture bladder with associated urethral injury.
- Multiple fracture involved bilateral superior & inferior pubic rami, right iliac crest abnormal widening pubic symphysis, also fracture involving the left iliac bone extending into left SI joint and left L5 transverse process.
- Large acute hematoma (30-60 HU) at upper to lower back areas, along posterior to erector spinae muscle, about 2.4 cm in maximal thickness, anterolateral aspect to right external oblique muscle and right quadratus lumborum. Minimal hematoma at left psoas muscle.
- Liver shows normal size, enhancement and normal contour without space taking lesion or intrahepatic bile duct dilatation.

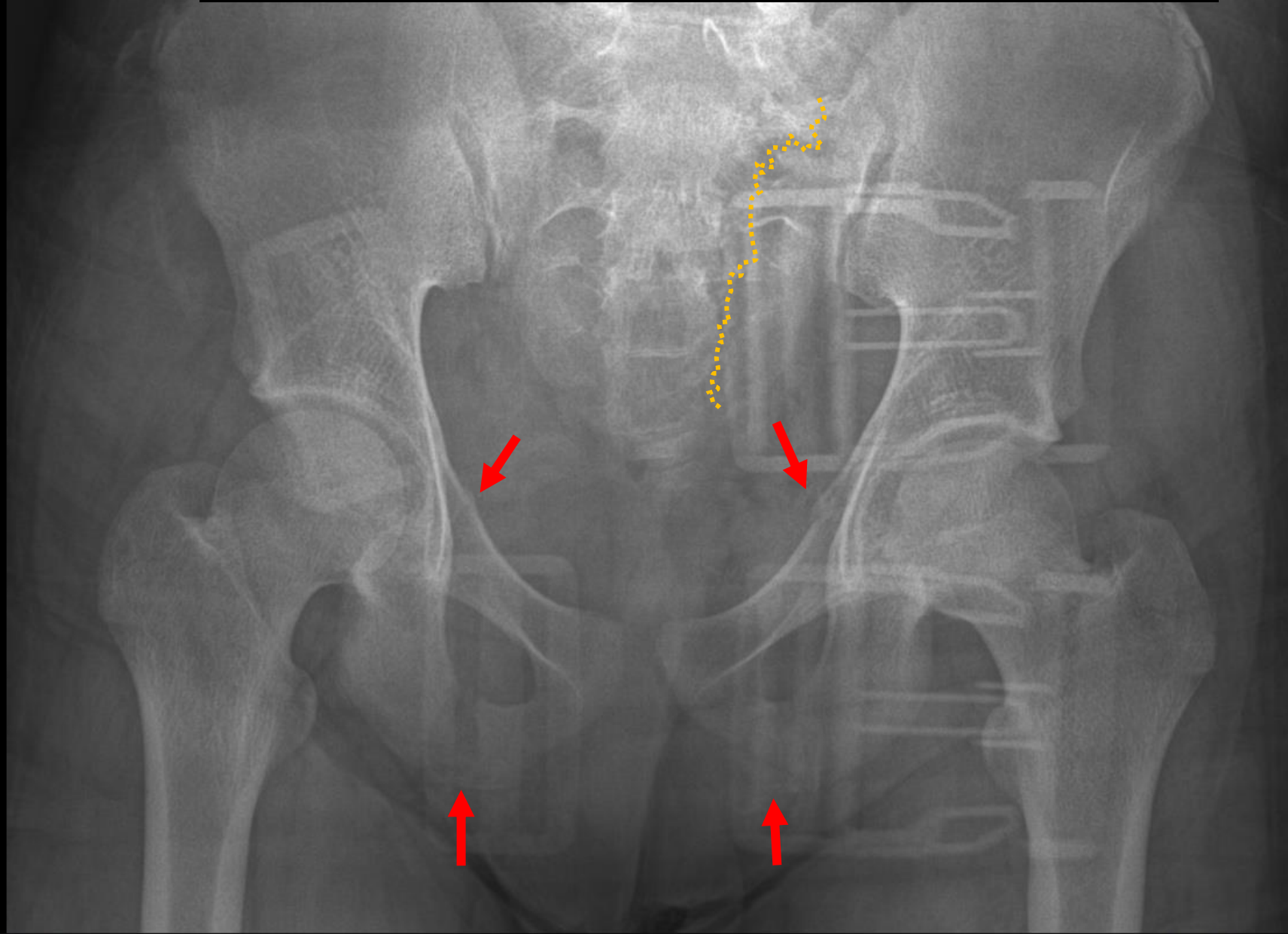
- There is no lymphadenopathy or ascites. Lung bases are clear. No pleural effusion.

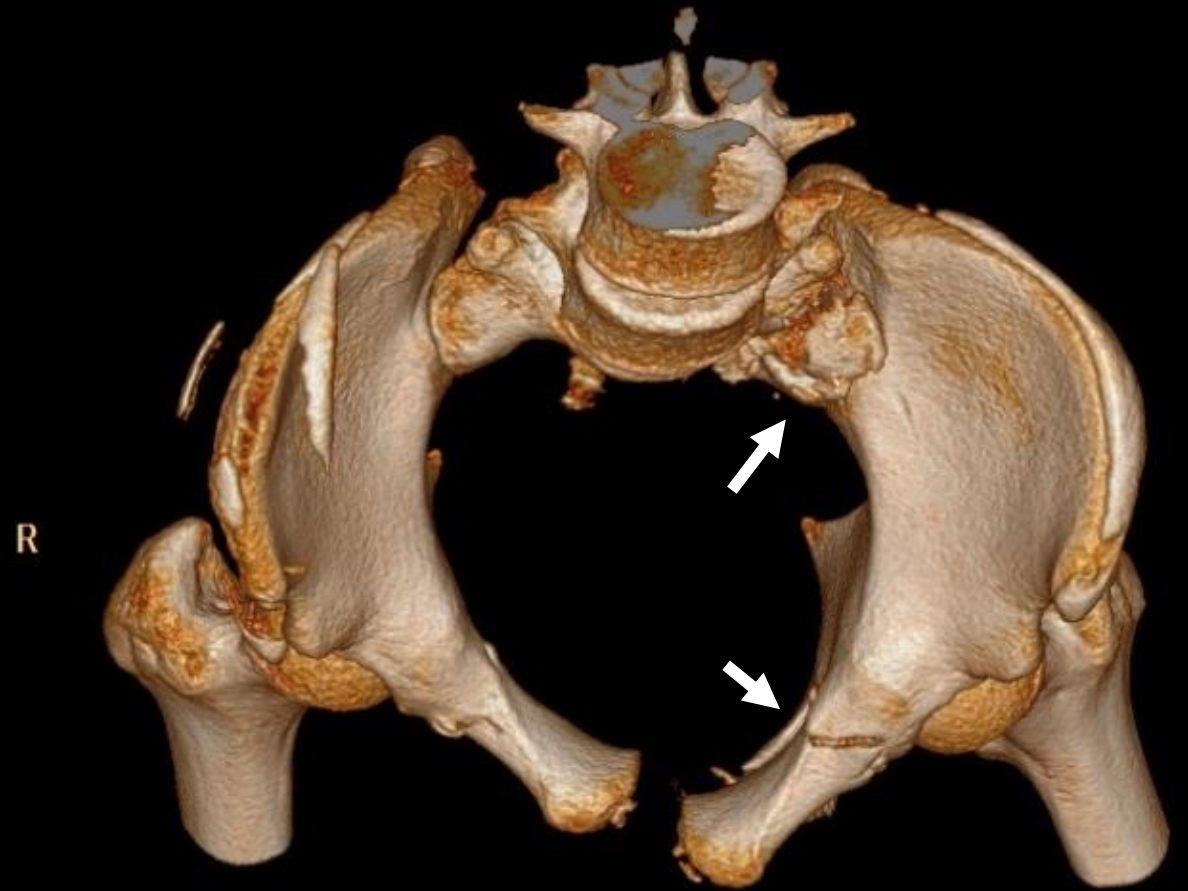
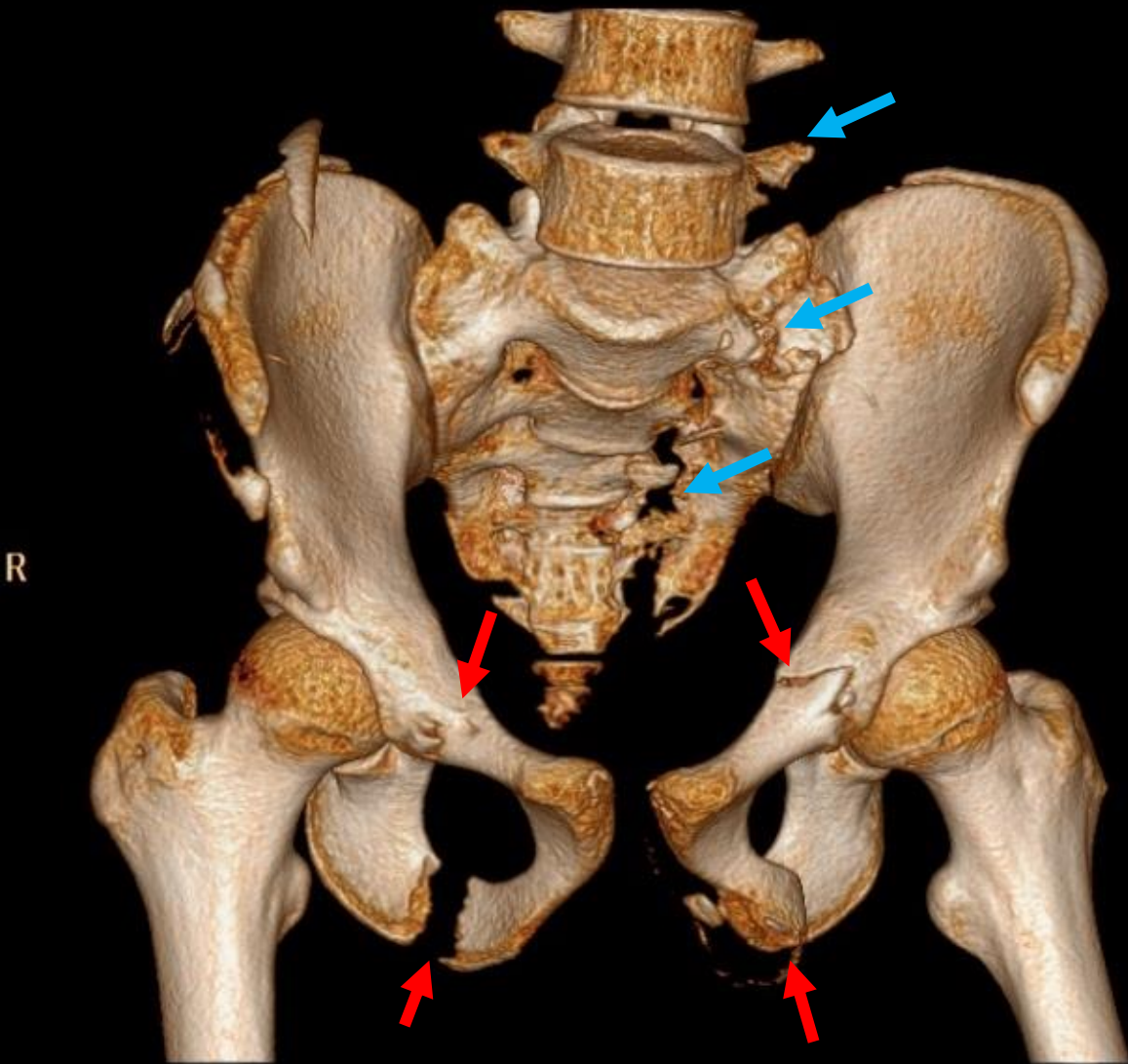
IMPRESSION:

- Evidence of moderate amount of free fluid throughout the lower abdominal and in pelvic cavity. Extraluminal contrast/ small free gas into perivesicle space and extravasation of urinary tract contrast media below the urogenital diaphragm and periurethral region and after delayed scan which increased extension of extraluminal contrast and abnormal position of foley's balloon catheter, suspicious of extraperitoneal rupture bladder with associated urethral injury.
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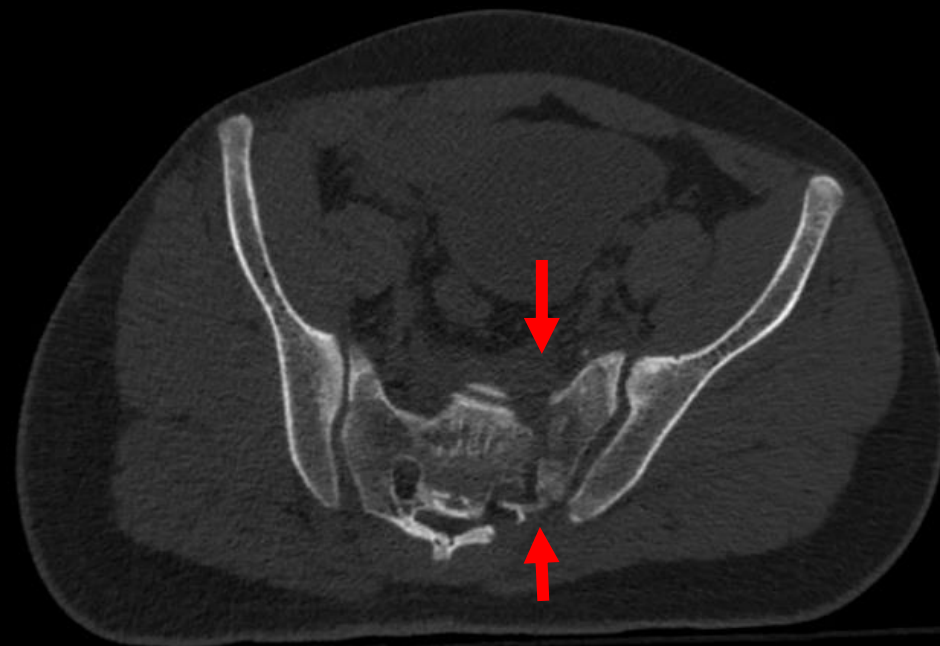
Straddle type pelvic fracture

* Always suspect for posterior ring fracture





Pelvic fracture: LC type
Rupture bladder with urethral injury
Initial hypovolemic shock with response to fluid resuscitation





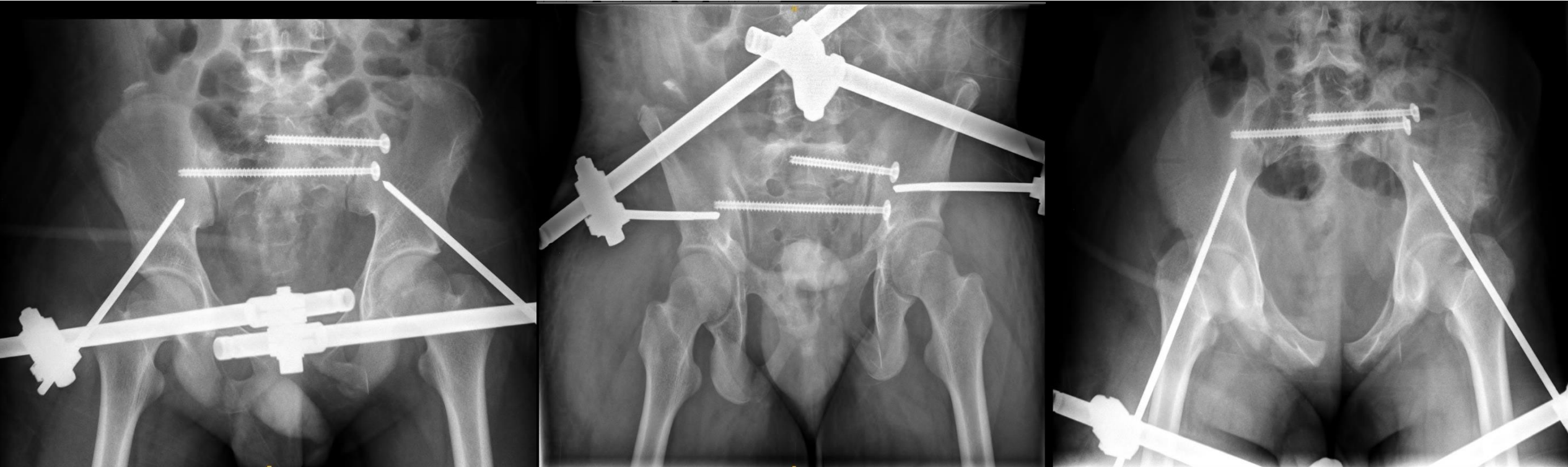
Suprapubic cystostomy



Double set up: intra-operative cytogram +/- explore laparotomy
Ortho → Application external fixator



Orthopedic definitive management



Percutaneous sacro-iliac screw screws fixation
With anterior external fixation



Conclusion

- Primary and secondary survey
- Remember fluid and blood component resuscitation
- More than 50% of pelvic fracture has associated injury
- Pelvic fracture bleeding from venous source 80%
- Arterial source need embolization or surgical pelvic packing
- Pelvic binder should apply initially in all suspected pelvic fracture



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Thank you

